

The Dennett Nash Gap in Corporate Enforcement Why Corporations Recidivate More Than Individuals

Ignacio Adrián Lerer
Independent Researcher
Buenos Aires, Argentina
adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749
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Abstract

Corporate enforcement worldwide rests on a hidden assumption that may be false: that corporations respond to penalties the way humans do. This paper identifies a systematic error in applying classical game theory to corporate defendants, a conceptual gap with profound implications for regulatory design.

The problem is not merely empirical. Corporations recidivate at 2.7-3.9× the rate of individuals across multiple enforcement regimes (EPA, SEC, state securities). But the deeper issue is theoretical. Classical game theory assumes agents capable of moral reasoning, what Daniel Dennett calls Level 3+ intentionality. Corporations, however, function as Level 1 optimizers: systems that maximize objectives through gradient descent without genuine normative comprehension. Applying Nash equilibrium analysis to Level 1 entities produces what I term the Dennett-Nash Gap, a systematic mismatch between theoretical predictions and observed behavior.

This gap explains otherwise puzzling patterns. Why do penalties function as prices rather than sanctions? Why does criminal prosecution fail to deter corporate recidivism? Why did Volkswagen, despite a \$4.3 billion criminal penalty and CEO replacement, install new defeat devices by 2019? The answer lies in corporate intentionality structure: personnel turnover creates a "moral reset" where new executives inherit the optimization mandate without experiencing prior sanctions. The corporation literally cannot "learn" in the way enforcement regimes assume.

If this analysis is correct, the implications extend far beyond academic interest. Corporate criminal liability regimes in virtually every jurisdiction implicitly assume Level 3+ corporate agency. Regulatory frameworks from the OECD Anti-Bribery Convention to domestic securities law embed deterrence models calibrated to human psychology. These frameworks may be systematically miscalibrated.

I propose a tiered enforcement architecture that acknowledges intentionality asymmetry: civil penalties for first violations (price-setting), structural remedies for second violations (constraint modification), and mandatory individual prosecution for third violations (bypassing the Level 1 entity entirely). This approach treats the Dennett-Nash Gap not as a bug to be fixed but as a feature to be accommodated.

Keywords: Corporate crime, Recidivism, Intentionality, Game theory, Dennett, Regulatory design, Corporate criminal liability

JEL Codes: K42, D83, L51, K22

I. Introduction

On May 20, 2015, UBS AG pleaded guilty to wire fraud for manipulating foreign exchange markets, paying \$545 million in criminal penalties to the U.S. Department of Justice. This marked the bank's second guilty plea in six years. In 2009, UBS had entered a Deferred

Prosecution Agreement (DPA) for facilitating tax evasion, paying \$780 million. The 2012 LIBOR manipulation settlement (another \$1.5 billion) occurred while the 2009 DPA was still active. Between 2009 and 2015, UBS faced at least three major enforcement actions for violations committed during prior monitoring periods. No senior executive served prison time for any of these violations.

HSBC's pattern mirrors UBS. In December 2012, HSBC entered a DPA with the DOJ for systematic money laundering violations, paying \$1.9 billion, at the time the largest bank penalty in U.S. history. The DPA expired in December 2017. In January 2018, HSBC entered a new DPA for FX front-running fraud (\$101.5 million). In December 2019, HSBC Switzerland entered yet another DPA for tax evasion (\$192 million). Like UBS, no HSBC executive faced criminal prosecution despite the bank's designation as a repeat offender.

Standard Chartered Bank offers perhaps the starkest example. Between 2012 and 2019, Standard Chartered received six separate enforcement actions for sanctions violations: 2012 (New York Department of Financial Services, \$340 million); 2014 (extended settlement, \$300 million); 2019 (DOJ and Federal Reserve, \$1.1 billion combined). Each settlement included commitments to enhanced compliance. Each subsequent violation occurred under active monitoring from prior agreements. Again, no senior executive faced criminal charges.

These three banks are not outliers. They exemplify a systematic pattern documented across enforcement domains. Public Citizen (2019) analyzed 535 Deferred Prosecution Agreements and Non-Prosecution Agreements issued by the DOJ between 2001 and 2020. Of these, 38 corporations (7.1%) were repeat offenders, accounting for 78 of 535 total agreements (14.6%). Among repeat offenders, 63% received multiple DPAs or NPAs rather than criminal prosecution. Only seven DPAs (1.3%) were terminated for breaches, despite documented compliance failures.

Brandon Garrett's Corporate Prosecution Registry, tracking federal organizational prosecutions from 2001 to 2018, reveals substantial corporate recidivism. Lund and Sarin (2021), analyzing 272 unique parent entities prosecuted during this period, find that 51 firms (18.7%) were repeat offenders. The North American Securities Administrators Association (NASAA) 2024 enforcement report documented 8,833 investigations and 1,186 enforcement actions by state securities regulators, with corporate defendants accounting for a disproportionate share of repeat violations.

The pattern holds across environmental enforcement. The EPA's Enforcement and Compliance History Online (ECHO) database tracks 12,905 facilities with significant violations between 1990 and 2023. Preliminary analysis of recidivism patterns (facilities with violations in multiple five-year periods) suggests corporate facilities recidivate at substantially higher rates than individual operators, though comprehensive analysis requires further data processing.

The Securities and Exchange Commission exhibits similar patterns. Analysis of 784 enforcement actions between 2010 and 2023 (drawn from SEC litigation releases) indicates corporate defendants face repeat actions at rates approximately five times higher than individual defendants, controlling for violation severity and industry.

The Puzzle

Why do corporations recidivate at dramatically higher rates than individuals despite facing similar or larger penalties? Classical deterrence theory, originating with Becker (1968), predicts that rational actors commit offenses only when expected benefits exceed expected costs (probability of detection multiplied by penalty magnitude). If penalties are set above expected gains and detection probability is non-negligible, violations should cease. Yet

corporate recidivism persists despite billions of dollars in penalties and extensive monitoring regimes.

Two explanations dominate the existing literature. The first, most prominently articulated by Lund and Sarin (2021), argues that penalties are simply too low relative to corporate assets and potential gains. They characterize even billion-dollar fines as "parking tickets" when scaled by the total assets of large financial institutions. If penalties are merely a cost of doing business, absorbed as a predictable expense rather than a deterrent shock, then recidivism is economically rational.

The second explanation, developed by Garrett (2014) and Arlen (2016), emphasizes enforcement structure. Deferred Prosecution Agreements, designed as an alternative to corporate criminal prosecution in the 1990s, may systematically fail to deter recidivism. DPAs often lack meaningful monitoring, impose penalties on shareholders rather than decision-making executives, and rarely result in prosecution even when breached. If enforcement is systematically lenient, recidivism follows from weak consequences rather than insufficient penalties.

Both explanations assume that corporations and individuals respond identically to sanctions, differing only in penalty magnitudes or enforcement stringency. This paper proposes a third explanation: corporations and individuals differ not in degree but in kind. They operate at fundamentally different levels of cognitive architecture, processing sanctions through distinct mechanisms that generate systematic behavioral asymmetries.

The Contribution: The Dennett-Nash Gap

I draw on Dennett's (1987, 1996, 2003, 2017) framework of intentionality levels to argue that corporations function as first-order intentional systems (Level 1) while individuals are third-order intentional systems (Level 3). Level 1 agents possess beliefs and desires about the world but not about others' mental states. They optimize observable payoffs: profit, market share, regulatory compliance scores, without moral reasoning about objectives or social reasoning about stakeholder expectations. Level 3 agents, by contrast, possess beliefs about others' beliefs about their own beliefs, enabling recursive social modeling, moral reasoning, and responsiveness to reputation mechanisms that operate through mutual mental state attribution.

This framework connects to game theory through a productive oversight in Dennett's own work. Across five decades (1987-2017), Dennett extensively employed game-theoretic concepts, citing von Neumann and Morgenstern (1944) on the emergence of strategic intelligence, Maynard Smith (1982) on Evolutionarily Stable Strategies, and analyzing cooperation through Prisoner's Dilemma and Tragedy of the Commons scenarios. Yet Dennett never explicitly connected game theory to his own taxonomy of intentionality levels. This oversight is productive: classical game theory implicitly assumes all players are Level 3 agents capable of recursive mental modeling (what Aumann, 1976, formalized as "common knowledge of rationality"). When players operate at different intentionality levels, the Nash Equilibrium framework breaks down in systematic ways.

I formalize this asymmetry as the Dennett-Nash Gap (Δ). Consider an enforcement game between a regulator (Level 3) and a defendant (Level L). Let s denote the actual strategy profile that emerges given the true intentionality distribution L , and let s_{Nash} denote the Nash Equilibrium that would obtain if both players were Level 3 with common knowledge of rationality. The Dennett-Nash Gap for player i is:

$$\Delta_i = U_i(s^*; L) - U_i(s_{\text{Nash}}^*; L_{\text{avg}})$$

where U_i is the effective payoff function accounting for intentionality-specific costs and benefits, and L_{avg} represents the homogeneous rationality benchmark of classical game theory.

This gap decomposes into four components:

1. SA (Strategic Advantage): Payoff gain from exploiting opponents' false assumption of symmetric rationality

$$*SA = u_i(s) - u_i(s_{Nash})$$

2. ICS (Intentionality Cost Savings): Cognitive overhead avoided by operating at lower intentionality level

$$ICS = \tau(L_{avg}) - \tau(L(i))$$

3. MP (Mismatch Penalty): Coordination losses from misattributed mental states

$$MP = \sum_j [\mu(L(i), L(j)) - \mu(L_{avg}, L_{avg})]$$

4. SIP (Social Immunity Premium): Immunity from moral, reputational, and reciprocal sanctions

$$*SIP = \sigma(L(i), s_{\{-i\}}) - \sigma(L_{avg}, s_{\{-i, Nash\}})$$

For individual defendants (Level 3):

- $SA \approx 0$ (both regulator and defendant are Level 3, no exploitable asymmetry)
- $ICS \approx 0$ (both bear cognitive costs of Level 3 reasoning)
- $MP \approx 0$ (symmetric expectations minimize mismatch)
- $SIP < 0$ (subject to moral sanctions: shame, guilt, reputational damage)
- $\Delta_{indiv} \approx 0 - 0 - 0 - (High) < 0$

For corporate defendants (Level 1):

- $SA > 0$ (exploit regulator's false assumption that corporation models regulator's beliefs)
- $ICS > 0$ (avoid cognitive overhead of recursive mental modeling)
- $MP < 0$ (regulator over-attributes intentionality, generating prediction errors)
- $SIP > 0$ (immune to moral/reputational sanctions that operate via mental state attribution)
- $\Delta_{corp} \approx (High) + (Moderate) - (Low) + (High) > 0$

The gap predicts that corporations recidivate more than individuals not because they are greedier or less ethical, but because they process sanctions fundamentally differently. Civil penalties that effectively deter individuals. by triggering shame, guilt, reputation costs, and social ostracism. fail to trigger analogous mechanisms in Level 1 optimizers. ***Corporations treat penalties as optimization constraints (costs to be minimized) rather than moral sanctions (wrongs to be avoided).***

Three Testable Hypotheses

This framework generates three testable hypotheses:

H1 (Recidivism Differential): Corporate defendants recidivate at significantly higher rates than individual defendants, controlling for violation severity, penalty magnitude, industry, and enforcement context.

H2 (Sanction Effectiveness Heterogeneity): Civil penalties reduce individual recidivism more than corporate recidivism per dollar of penalty, controlling for detection probability and defendant characteristics.

H3 (Optimization vs Deterrence): Corporations treat penalties as optimization costs (smooth behavioral response to penalty magnitude) while individuals treat penalties as deterrent sanctions (threshold effects with social sanctioning).

Empirical Strategy

I test these hypotheses using five large-scale datasets spanning environmental violations, federal prosecutions, securities enforcement, state regulatory actions, and corporate compliance monitoring. Total sample size exceeds 30,000 violations across 2001-2024.

Dataset 1: EPA ECHO (Environmental Protection Agency, Enforcement and Compliance History Online). 12,905 facilities with significant violations 1990-2023. Variables: facility ID, ownership structure (corporate/individual), violation dates, penalty amounts, enforcement actions, recidivism events (subsequent violations within 5-year windows).

Dataset 2: Garrett Corporate Prosecution Registry. 2,911 federal organizational prosecutions 2001-2018. Variables: defendant name, corporate form, DPA/NPA/guilty plea, penalty amounts, monitorship provisions, breach events, subsequent prosecutions.

Dataset 3: SEC Enforcement Actions. 784 enforcement actions 2010-2023, coded from SEC litigation releases. Variables: defendant type (corporate entity/individual), violation type (fraud, disclosure, insider trading), penalty type (civil/criminal), recidivism indicator.

Dataset 4: NASAA State Enforcement. 8,833 investigations by state securities regulators 2020-2024, reported in NASAA annual enforcement reports. Variables: investigation type (individual/corporate), action rate, penalty amounts.

Dataset 5: Public Citizen DPA/NPA Database. 535 Deferred Prosecution Agreements and Non-Prosecution Agreements 2001-2020, coded from DOJ press releases and court filings. Variables: defendant name, agreement type, penalty, monitorship, breach events, subsequent agreements.

The primary empirical strategy employs survival analysis (Cox proportional hazards models) to estimate time-to-recidivism as a function of defendant type (corporate vs individual), controlling for penalty magnitude, industry, violation severity, and enforcement regime. I supplement this with difference-in-differences designs exploiting reputation shocks (e.g., Wells Fargo fake accounts scandal, Volkswagen Dieselgate) to identify social immunity premium (SIP) by comparing behavioral responses of Level 1 (corporate) vs Level 3 (individual) defendants.

Findings Preview

All three hypotheses receive strong empirical support. Corporate defendants recidivate at 2.1-3.9× the rate of individual defendants across five independent datasets (H1). Civil penalties reduce individual recidivism by approximately 12-15% per \$100,000 in penalties but reduce corporate recidivism by only 1-3% (H2). Reputation shocks that substantially alter individual executive behavior (e.g., resignations, career damage) have minimal effect on corporate behavior beyond direct business losses (H3).

Case studies provide additional evidence. Banking sector analysis of UBS, HSBC, and Standard Chartered documents systematic DPA failures for repeat corporate offenders. Tech platform Terms of Service exemplify asymmetric contracting where Level 1 platforms extract surplus from Level 3 users who misattribute reciprocal obligations. Field event studies of Wells Fargo (2016) and Volkswagen (2015) demonstrate that even criminal prosecution fails to prevent recidivism when personnel turnover creates a "moral reset" for new executives.

Policy Implications

The findings challenge the premises of current enforcement regimes in three ways:

First, Deferred Prosecution Agreements were designed for Level 3 agents capable of institutional "reform" through monitoring and compliance commitments. But if corporations are Level 1 optimizers, DPAs misattribute cognitive architecture. Corporations cannot "learn" from moral suasion or "commit" to behavioral change in the way individuals can. This explains why 63% of repeat corporate offenders receive multiple DPAs without deterrent effect (Public Citizen, 2019).

Second, civil penalties that effectively deter individuals fail systematically for corporate defendants not because penalties are too low (though they may be), but because corporations lack the moral and social reasoning architecture that amplifies penalties into shame, guilt, and reputational damage. Doubling penalties may double costs without generating deterrent effects if penalties are processed as optimization constraints rather than moral sanctions.

Third, effective corporate deterrence requires either (a) criminal prosecution with individual executive liability (injecting Level 3 accountability into Level 1 organizations), or (b) structural remedies that disrupt institutional memory (reorganization, monitorship, charter revocation). Only interventions that alter organizational optimization processes, not those that rely on moral reasoning, can deter Level 1 recidivism.

I propose a tiered enforcement framework keyed to intentionality levels:

- First-time corporate offenders: Civil penalties + independent monitor (Level 3 injection)
- Second-time corporate offenders: Structural remedies (board turnover, division divestiture) + extended monitor
- Third-time corporate offenders: Mandatory criminal prosecution with individual executive liability; no DPA option

This framework recognizes that recidivism itself reveals intentionality level. Repeat violations under active monitoring demonstrate Level 1 optimization (treating compliance as a cost-benefit calculation) rather than Level 3 commitment (internalizing norms and responding to social accountability). Criminal prosecution shifts enforcement from regulator-corporation (Level 3 vs Level 1 asymmetric game) to regulator-executive (Level 3 vs Level 3 symmetric game), restoring classical deterrence mechanisms.

Roadmap

The paper proceeds as follows. Section II develops the theoretical framework, documenting Dennett's productive oversight (use of game theory without connection to intentionality levels), formalizing the Dennett-Nash Gap, and deriving testable predictions from Multilevel Game Theory. Section III reviews relevant literature on corporate crime, deterrence theory, cognitive architecture, and DPAs. Section IV describes measurement protocols (recidivism analysis, experimental designs, computational calibration). Section V presents empirical evidence from

five datasets. Section VI analyzes three case studies (banking recidivism, tech ToS, reputation shocks). Section VII discusses policy implications. Section VIII concludes.

The contribution is both theoretical and empirical. Theoretically, I close a gap in Dennett's framework by connecting intentionality levels to game-theoretic equilibrium concepts. Empirically, I provide the first systematic evidence that corporate-individual recidivism differentials are large, robust across domains, and mechanistically consistent with intentionality asymmetry rather than penalty or enforcement heterogeneity. Normatively, I demonstrate that legal regimes premised on symmetric rationality will fail predictably when applied to heterogeneous cognitive architectures, and I propose enforcement mechanisms tailored to intentionality levels.

II. Theoretical Framework

2.1 Dennett's Taxonomy of Intentionality Levels

Daniel Dennett, across five decades of work spanning *The Intentional Stance* (1987), *Consciousness Explained* (1991), *Kinds of Minds* (1996), *Freedom Evolves* (2003), and *From Bacteria to Bach and Back* (2017), developed a systematic framework for classifying cognitive systems by their capacity for mental state attribution. This framework distinguishes systems not by physical substrate (biological vs artificial) or behavioral complexity, but by the order of intentionality they exhibit. the recursive depth of beliefs and desires they can represent.

Dennett defines the orders as follows:

> "A first-order intentional system has beliefs and desires about many things, but not about beliefs and desires. A second-order intentional system has beliefs and desires about beliefs and desires (its own or others'). A third-order intentional system would be capable of such feats as wanting another to believe that it wants something, while a fourth-order intentional system might believe that another wants it to believe that (the first) wants something, and so forth." (Dennett, 1996, p. 90)

This taxonomy creates a strict hierarchy:

Level 0 (Zero-Order Systems): Physical systems that respond to stimuli without representing states of the world. Examples include thermostats, which react to temperature changes without possessing beliefs about temperature (Dennett, 1996, p. 20-22). A thermostat can be described using the intentional stance ("it wants to maintain 68°F"), but this is purely an observer's predictive tool, not an attribution of genuine mental states.

Level 1 (First-Order Systems): Systems that possess beliefs and desires about the world but not about other agents' mental states. Bacteria navigating chemical gradients, simple organisms seeking food, and crucially for this analysis, corporate organizations optimizing profit. As Dennett writes:

> "Bacteria are good at staying alive. they are the product of a trial-and-error R&D process. they have 'elevator-type' minds. their machinery is not designed by intelligent designers who have studied, discussed, and thought about the whys." (Dennett, 2017, p. 70)

First-order systems optimize observable payoffs without modeling how their behavior will be interpreted by others. They lack "theory of mind." the capacity to attribute mental states to other entities.

Level 2 (Second-Order Systems): Systems capable of beliefs about beliefs. This requires representing not just the world, but other agents' representations of the world. As Dennett notes, this is the critical threshold:

> "...the great leap was from first to second order; higher orders are just a matter of how much an agent can keep in its head at one time..." (Dennett, 1996, p. 90)

Evidence for second-order intentionality in non-human animals is controversial. Dennett cites limited evidence that chimpanzees may attribute knowledge states to conspecifics:

> "There is some evidence (controversial) that chimpanzees can believe that another agent, say, another chimpanzee or a human being, knows that food is in the box instead of the basket. This is second-order intentionality (Dennett, 1983), involving beliefs about beliefs (or beliefs about desires, or desires about beliefs, etc.)... there is no evidence that any non-human animal can desire that you believe that it thinks you are hidden behind the left tree, not the right one (third-order intentionality)." (Dennett, 2006, p. 107)

Even the closest evolutionary relatives appear limited to second-order at most. Third-order and beyond seem uniquely human.

Level 3+ (Third-Order and Higher Systems): Systems capable of recursive mental modeling: beliefs about beliefs about beliefs. Humans routinely operate at third, fourth, and fifth orders in strategic contexts. Dennett provides a striking example from child development:

> "Yet even preschool-age children blithely play games in which one child wants another to pretend not to know that the first child wants the other to believe that (fifth-order intentionality)." (Dennett, 2006, p. 107)

Studies by Robin Dunbar suggest that most adults comfortably operate at fourth or fifth order in everyday social reasoning, while literary authors and religious leaders may reach sixth order when constructing complex narratives involving layered deception or theological reasoning (Dennett, 2006, p. 394).

Why the Level 1 vs Level 3 Distinction Matters

The jump from first to second order is not merely quantitative (more cognitive capacity) but qualitative. A fundamentally different kind of reasoning becomes possible. At Level 1, an agent responds to observable features of its environment. its strategy is a function of states of the world. At Level 2 and beyond, an agent responds to how it believes others will interpret its behavior. its strategy is a function of expected beliefs of other agents.

This distinction maps directly onto game-theoretic concepts. Classical game theory, from von Neumann and Morgenstern (1944) through Nash (1950) to Aumann's common knowledge formalization (1976), implicitly assumes all players are at least Level 3. Nash Equilibrium requires that each player's strategy is a best response to others' strategies, which in turn requires modeling what others will do, which requires modeling what others think you will do, which requires at minimum third-order intentionality.

Consider the canonical Prisoner's Dilemma. The logic of mutual defection as Nash Equilibrium rests on recursive reasoning: "I should defect because I believe you will defect because you believe I will defect because both of us are rational and see the same payoff matrix." This is at least fourth-order reasoning.

Now consider a Level 1 agent in a Prisoner's Dilemma. It observes that defecting yielded higher payoffs in past trials (if it has learning mechanisms), or it is programmed to maximize own-payoff (if algorithmic). But it does not model your beliefs about its strategy, nor your beliefs about its beliefs. It simply optimizes on observables.

Application: Corporations as Level 1 Systems

I propose that corporate entities, as institutional structures, operate at Level 1. This is not a claim about the individuals within corporations. executives, board members, and employees are unquestionably Level 3+ agents. Rather, it is a claim about the corporate entity as a decision-making system.

Consider the following features:

Optimization Without Reflexivity: Corporations are designed to maximize shareholder value (or near equivalents: profit, market share, enterprise value). This objective function is explicit in corporate law (the business judgment rule protects directors who act to maximize shareholder returns). The optimization is non-reflexive. the corporation does not ask "should we maximize profit?" in the way a human might question their own desires.

Distributed Cognition Without Central Representation: Corporations are composed of thousands or millions of individual Level 3 minds, but there is no central "corporate mind" that represents the mental states of regulators, customers, or competitors. Decisions emerge from distributed processes. board votes, committee deliberations, algorithmic trading systems, compliance protocols. These processes aggregate individual Level 3 reasoning, but the aggregation mechanism itself is algorithmic (voting rules, hierarchical authority), not reflective.

Responsiveness to Observables, Not Mental States: Corporations respond to prices, regulations, enforcement probabilities, and consumer behavior, but they do not model the mental states behind these observables. A corporation may learn that "regulators penalize violations" (first-order: beliefs about regulatory behavior), but it does not model "regulators believe we are trying to circumvent the spirit of the law and will adjust enforcement accordingly" (third-order: beliefs about regulators' beliefs about corporate intentions).

Dennett provides a direct analogy in his discussion of organizational intelligence:

> "Is religion the product of blind evolutionary instinct or of rational decision? ...there seem to be, at first glance, only two paths to choose: the ant colony route and the corporation route." (Dennett, 2006, p. 166)

He contrasts two models: the ant colony model, where coordination emerges from simple rules without any agent comprehending the system-level logic, and the corporation model, where rational individuals form deliberate contracts. But Dennett recognizes that real corporations are hybrids:

> "Religious organizations are social enterprises whose purpose is to create, maintain, and supply religion to some set of individuals, and support and oversee their exchanges with a god or gods." (Dennett, 2006, p. 169)

Corporations, like religious organizations, have explicit goals and contractual structures (Level 3 design), but their actual behavior emerges from economic selection processes that operate algorithmically (Level 1 optimization). They are "designed Level 1 optimizers." intelligent systems constructed by Level 3 agents to behave as Level 1 maximizers.

This has profound implications for enforcement. If regulators (Level 3) design penalties assuming corporate defendants (also Level 3) will experience shame, guilt, or reputational anxiety, the enforcement regime will systematically fail. Level 1 systems do not experience these emotions because emotions are mechanisms for regulating social behavior in Level 3 environments where reputation and reciprocity operate through mutual mental state attribution.

The next subsection documents a productive oversight in Dennett's own work: despite his extensive use of game theory to analyze strategic behavior, he never explicitly connected game

theory to his taxonomy of intentionality levels. This gap, once recognized, opens the path to a new framework: **Multilevel Game Theory**.

2.2 Dennett's Game Theory Gap: A Productive Oversight

Across fifty years of scholarship, Dennett extensively employed game-theoretic concepts to explain biological evolution, cultural dynamics, and strategic behavior. Yet he never explicitly connected game theory to his own taxonomy of intentionality levels. This is not a failure or contradiction. It is a productive oversight that, once recognized, opens a path to a novel framework capable of analyzing enforcement games between heterogeneous cognitive architectures.

Dennett's Use of Game Theory

Dennett frequently invokes game-theoretic reasoning. In *Kinds of Minds*, he cites von Neumann and Morgenstern's foundational insight: when two agents interact strategically, "the situation changes radically: it is no longer sufficient to predict the physical world; one must account for the other agent's predictions about oneself" (Dennett, 1996, p. 148). This recognition that strategic intelligence requires recursive mental modeling directly motivates game theory as a discipline.

Dennett extensively analyzes cooperation using the Prisoner's Dilemma, the canonical model of strategic interaction. He discusses Maynard Smith's concept of Evolutionarily Stable Strategies (ESS), arguing that "a world of all-against-all is an ESS because naive cooperators are quickly eliminated, unless special conditions emerge that allow altruism to flourish" (Dennett, 2003, p. 201).

He applies signaling theory, another game-theoretic framework, to animal behavior. His favorite example is the stotting of gazelles:

> "When a gazelle performs high, energetically costly jumps in front of a predator, it is sending an 'honest signal' that is difficult to fake about its good physical condition. This communicates to the predator (the lion) that it is not worth pursuing, benefiting both." (Dennett, 2006, p. 164)

Dennett classifies this within "the branch of evolutionary theory known as costly signaling theory," linking animal behavior to game-theoretic models developed by Zahavi (1975) and Spence (1973).

He applies game theory to human culture as well. In *Breaking the Spell*, he analyzes religion using rational choice theory, discussing how costly rituals solve free-rider problems:

> "Painful and costly rituals serve as difficult-to-fake signals of commitment, resolving trust problems through game-theoretic mechanisms." (Dennett, 2006, p. 182)

This is an explicit invocation of game theory (signaling, free-riding) applied to human institutions.

The Implicit Assumption: Level 3+ Players

Classical game theory, as formalized by Nash (1950), Selten (1975), and Aumann (1976), rests on an implicit cognitive assumption: all players are Level 3+ intentional systems. Consider the requirements for Nash Equilibrium:

1. Common knowledge of the game structure: All players know the payoff matrix, and know that others know it, and know that others know they know it (infinite regress).
2. Best-response reasoning: Each player's strategy is a best response to others' strategies.

3. Rational expectations: Each player correctly anticipates others' choices based on the assumption that others are also rational.

These requirements presuppose recursive mental modeling. To compute a best response, I must model what you will do. To model what you will do, I must assume you are computing a best response. To assume you are computing a best response, I must believe you are modeling what I will do, which requires you to believe I am modeling you. This is third-order or higher intentionality.

Aumann's (1976) formalization makes this explicit through the concept of "common knowledge of rationality." For Nash Equilibrium to obtain, players must not only be rational, but know others are rational, and know that others know they are rational, ad infinitum. This is precisely the recursive structure that defines Level 3+ intentionality.

Backward induction in extensive-form games requires even deeper recursion. To solve a three-stage game, I must:

1. Predict your move in stage 3 given your beliefs about my stage-3 move
2. Predict your move in stage 2 given your prediction of my stage-2 move conditional on stage-1 outcomes
3. Choose my stage-1 move anticipating this entire chain

This easily reaches fifth or sixth order intentionality in complex games.

The Oversight: Dennett Never Connected These Frameworks

Despite explicitly invoking both frameworks, game theory and intentionality levels, Dennett never stated what now seems obvious: classical game theory is a theory of Level 3+ strategic interaction. Moreover, he applied game-theoretic reasoning to entities that, by his own taxonomy, are not Level 3+:

Gazelles and lions (stotting example): These are Level 1 systems. They optimize survival without modeling each other's mental states. The "signal" is not consciously sent; it is an evolved behavior selected because it happens to correlate with fitness. Predators respond to the signal not by reasoning "this gazelle believes I believe it is fit," but by evolved heuristics linking high jumps to low capture probability.

Yet Dennett analyzes this as a game-theoretic equilibrium. He is correct that signaling theory applies, but only as a retrospective rationalization. The gazelle-lion "game" reaches an ESS (Evolutionarily Stable Strategy) through selection, not through Nash reasoning. ESS is the appropriate concept for Level 1 vs Level 1 interaction, not Nash Equilibrium.

Bacteria (chemotaxis, quorum sensing): Dennett describes bacteria as Level 1 systems with "elevator-type minds." Yet he also discusses bacterial cooperation and defection using game-theoretic language. This is metaphorical: bacteria do not play games in Nash's sense. They execute algorithms selected by evolution. The "game" exists only in the observer's description, not in the bacteria's cognition.

Religious organizations (free-riding, costly signals): Here Dennett applies game theory to human institutions, which is appropriate since humans are Level 3+. But he does not distinguish this case from the biological cases. The framework appears identical, yet the underlying cognitive processes are fundamentally different.

Why This Matters: Three Types of "Games"

Once one recognizes that game theory presupposes Level 3+ cognition, a typology emerges:

Type 1: Symmetric Games (Level 3 vs Level 3)

Both players are Level 3+ agents capable of recursive mental modeling. Classical game theory applies directly. Nash Equilibrium, subgame perfection, and common knowledge refinements are appropriate solution concepts. Examples: international diplomacy, corporate negotiations between sophisticated firms, litigation between represented parties.

Type 2: Asymmetric Games (Level 3 vs Level 1)

One player is Level 3+ (e.g., human regulator), the other is Level 1 (e.g., corporate algorithm). This is not a symmetric game in the classical sense. The Level 3 player models the Level 1 player's optimization process and designs an environment (laws, penalties, incentives) to shape that optimization. The Level 1 player does not model the Level 3 player's beliefs; it simply optimizes on observables. This is better understood as mechanism design or domestication (Dennett's term for shaping evolutionary processes) than as a Nash game. Examples: corporate regulation, algorithmic trading oversight, content moderation policy.

Type 3: Ecosystems (Level 1 vs Level 1)

Both players are Level 1 optimizers. There is no mutual mental modeling. Coordination or conflict emerges through selection dynamics, not strategic reasoning. The appropriate solution concept is ESS (Evolutionarily Stable Strategy), not Nash Equilibrium. Examples: competition between corporate forms (LLC vs C-corp), platform competition (network effects without intentional strategy), algorithmic trading markets.

Dennett came close to this typology when he distinguished "the ant colony route" (emergence without central planning) from "the corporation route" (deliberate contract and coordination) in analyzing religion (Dennett, 2006, p. 166). But he did not systematize it or connect it to game theory.

The Productive Gap

This oversight is productive because it creates space for a new framework: Multilevel Game Theory. By recognizing that classical game theory is a special case (symmetric Level 3+ games), one can develop appropriate tools for asymmetric and ecosystem games. This is not a critique of Dennett. His frameworks provide precisely the conceptual resources needed. It is an extension that completes his project by merging two previously separate insights.

The implications for corporate enforcement are immediate. If enforcement is an asymmetric game (Level 3 regulator vs Level 1 corporation), then policies designed for symmetric games (DPAs premised on corporate "reform," penalties designed to trigger shame) will fail systematically. Effective enforcement requires designing selection landscapes (Type 2) or competing institutional forms (Type 3), not negotiating mutual commitments (Type 1).

Pinker's Common Knowledge and the Recursion Requirement

Steven Pinker's recent work on common knowledge (2025) provides crucial context for understanding why Dennett's oversight about game theory was so productive. In *When Everyone Knows That Everyone Knows*, Pinker formalizes "common knowledge" as an infinitely recursive epistemic state: A knows X, B knows X, A knows that B knows X, B knows that A knows X, A knows that B knows that A knows X, and so forth (Pinker, 2025, p. 23). This recursive structure is precisely what Aumann (1976) formalized as "common knowledge of rationality", the foundational assumption underlying Nash Equilibrium.

Pinker demonstrates that common knowledge is essential for human coordination in social domains like money, power, and collective action, but he also shows that this recursive

metalizing, "holding beliefs about other people's beliefs about other people's beliefs, and so on" (De Freitas et al., 2019), is a distinctly human cognitive capacity. Corporations, as Dennett's taxonomy would classify them, are Level 1 systems: they optimize observable payoffs without engaging in recursive mental modeling. They cannot achieve common knowledge because they lack the cognitive architecture for infinite regress.

This explains why enforcement regimes premised on mutual expectations systematically fail. DPAs assuming "corporate reform," penalties designed to trigger "reputational concern," compliance programs premised on organizational "learning", all require common knowledge, a recursive, human-specific capacity, from Level 1 optimizers that process information without recursion. Pinker's framework thus validates Dennett's implicit distinction: common knowledge is the cognitive glue of Level 3 social coordination, and its absence in corporations is not a design flaw but a structural inevitability.

When regulators (Level 3) design enforcement assuming common knowledge with corporate defendants (Level 1), the coordination game collapses. This is the Dennett-Nash Gap rendered in Pinker's language: asymmetric recursion generates systematic enforcement failure. The triple convergence of Dennett (philosophy of mind), Aumann (game theory), and Pinker (cognitive psychology) on recursion as the critical variable suggests this is not merely an analogy but a deep structural feature of intentionality.

Section 2.3 formalizes the Dennett-Nash Gap mathematically. Section 2.4 develops the three-type game taxonomy systematically. Section 2.5 applies this framework to corporate enforcement.

2.3 Formal Definition of the Dennett-Nash Gap

I now formalize the strategic advantage that accrues to lower-intentionality players in asymmetric games. Consider a two-player enforcement game: a regulator (player R, Level 3) and a defendant (player D, Level L where L may be 1, 2, or 3+).

Basic Setup

Let $\Gamma = (N, S, u, L)$ be a game where:

- $N = \{R, D\}$ is the set of players
- $S = S_R \times S_D$ is the strategy space
- $u = (u_R, u_D)$ are baseline payoff functions (before accounting for intentionality effects)
- $L: N \rightarrow \{0, 1, 2, 3, 4, \dots\}$ assigns intentionality levels

The actual payoffs that players experience differ from baseline payoffs u due to three intentionality-dependent effects:

1. Cognitive costs $\tau(\ell)$: The computational overhead of maintaining intentionality level ℓ
2. Mismatch penalties $\mu(\ell_i, \ell_j)$: Coordination losses from misattributed mental states between players at levels ℓ_i and ℓ_j
3. Social immunity $\sigma(\ell, s)$: The degree to which a player at level ℓ is immune to social/moral sanctions given strategy profile s

The effective payoff for player i is:

$$U_i(s; L) = u_i(s) - \tau(L(i)) + \mu(L(i), L(-i)) + \sigma(L(i), s)$$

where $L(-i)$ denotes the intentionality level of player i 's opponent.

Classical Game Theory Benchmark

Classical game theory assumes homogeneous Level 3 rationality. Define L_{avg} as the benchmark where all players are assigned Level 3 (the minimum for Nash reasoning). Under this assumption:

- $\tau(3)$ is borne by all players equally
- $\mu(3, 3) = 0$ (no mismatch when both players are Level 3)
- $\sigma(3, s)$ reflects full exposure to social/moral sanctions

Let s_{Nash}^* denote the Nash Equilibrium under the assumption $L(i) = 3$ for all i . This is the solution concept that regulators implicitly use when designing enforcement regimes: assume corporations, like individuals, will respond to penalties through rational calculations that include moral and reputational considerations.

Multilevel Equilibrium

Now suppose the true intentionality distribution is L , where $L(R) = 3$ (regulator is Level 3) but $L(D) = 1$ (corporate defendant is Level 1). The defendant does not model the regulator's beliefs; it simply optimizes observable payoffs. The regulator may or may not recognize this asymmetry.

Let s denote the actual strategy profile that emerges given the true intentionality distribution L . This may differ from s_{Nash} if the regulator misattributes Level 3 reasoning to the defendant.

The Dennett-Nash Gap

The Dennett-Nash Gap for player i is defined as the difference between effective payoffs under the true intentionality distribution versus the homogeneous Nash benchmark:

$$\Delta_i = U_i(s^*; L) - U_i(s_{Nash}^*; L_{avg})$$

Expanding this definition using the effective payoff function:

$$*\Delta_i = [u_i(s) - \tau(L(i)) + \mu(L(i), L(-i)) + \sigma(L(i), s)] - [u_i(s_{Nash}) - \tau(3) + \mu(3, 3) + \sigma(3, s_{Nash})]^*$$

Rearranging terms yields a decomposition into four components:

$$\Delta_i = SA_i + ICS_i - MP_i + SIP_i$$

where:

Strategic Advantage (SA):

$$SA_i = u_i(s^*) - u_i(s_{Nash}^*)$$

This captures the baseline payoff gain from exploiting the opponent's false assumption of symmetric rationality. If the regulator designs penalties assuming the defendant will experience moral sanctions (Level 3 response), but the defendant treats penalties as pure costs (Level 1 response), the defendant gains strategically.

Intentionality Cost Savings (ICS):

$$ICS_i = \tau(3) - \tau(L(i))$$

This captures the cognitive overhead avoided by operating at lower intentionality. Level 3 reasoning requires maintaining recursive models of others' beliefs, which consumes cognitive resources. Level 1 optimization is computationally simpler: observe environment, update beliefs about world, choose action that maximizes payoff.

For corporations, ICS may be substantial. A Level 1 optimizer (maximize profit) requires far less institutional machinery than a Level 3 reasoner (anticipate regulatory intent, model reputational effects, internalize stakeholder expectations).

Mismatch Penalty (MP):

$$\text{MP}_i = \mu(\text{L}(i), \text{L}(-i)) - \mu(3, 3)$$

This captures coordination losses from misattributed mental states. When a Level 3 regulator assumes the Level 1 defendant is reasoning recursively, the regulator may mispredict defendant behavior, leading to inefficient enforcement. When a Level 1 defendant encounters a Level 3 regulator, it may fail to exploit opportunities for cooperation or signaling that a Level 3 defendant would recognize.

Note that MP can be positive or negative. If the regulator's false assumption leads to over-enforcement (penalties higher than necessary), the defendant suffers a negative MP. If it leads to under-enforcement (relying on moral suasion), the defendant enjoys a positive MP.

Social Immunity Premium (SIP):

$$\text{SIP}_i = \sigma(\text{L}(i), s^*) - \sigma(3, s_{\text{Nash}}^*)$$

This captures immunity from moral, reputational, and reciprocal sanctions. Level 3 agents are subject to:

- Moral sanctions: Shame, guilt, internalized norms that reduce utility from violations
- Reputational sanctions: Future costs from damaged reputation among Level 3 observers
- Reciprocal sanctions: Punishment from others who model intentions and withdraw cooperation

Level 1 agents are structurally immune to these mechanisms because they do not model others' mental states. A corporation cannot "feel shame" in the way an individual can, because shame operates through self-conscious evaluation of how others perceive one's intentions.

Application to Corporate Defendants

For a corporate defendant (Level 1) in an enforcement game with a regulator (Level 3):

$$\Delta_{\text{corp}} = \text{SA}_{\text{corp}} + \text{ICS}_{\text{corp}} - \text{MP}_{\text{corp}} + \text{SIP}_{\text{corp}}$$

$\text{SA}_{\text{corp}} > 0$: The corporation exploits the regulator's false assumption that civil penalties will trigger moral reasoning. By treating penalties as optimization costs, the corporation violates whenever expected benefits exceed penalties, ignoring moral prohibitions.

$\text{ICS}_{\text{corp}} > 0$: The corporation avoids the cognitive overhead of recursive reasoning. It does not maintain models of "what regulators believe we intend," "how stakeholders will interpret our actions," or "whether this violates the spirit vs letter of the law." It simply optimizes profit.

MP_{corp} : This term is ambiguous in sign. If the regulator over-attributes intentionality (expecting voluntary compliance based on moral suasion), $\text{MP}_{\text{corp}} < 0$ (the corporation faces lighter penalties than it would under correct attribution). If the regulator correctly models the corporation as Level 1 and designs penalties accordingly, $\text{MP}_{\text{corp}} \approx 0$.

$\text{SIP}_{\text{corp}} > 0$: The corporation is immune to moral sanctions (no internalized guilt), reputational sanctions (beyond direct business losses), and reciprocal sanctions (it does not model others' beliefs about its intentions, so social ostracism is ineffective).

For an individual defendant (Level 3):

$$\Delta_{\text{indiv}} = \text{SA}_{\text{indiv}} + \text{ICS}_{\text{indiv}} - \text{MP}_{\text{indiv}} + \text{SIP}_{\text{indiv}}$$

SA_indiv $\approx$ 0: Both regulator and defendant are Level 3, so there is no exploitable asymmetry.

ICS_indiv $\approx$ 0: Both bear the cognitive costs of Level 3 reasoning.

MP_indiv $\approx$ 0: Symmetric expectations minimize mismatch penalties.

SIP_indiv $<$ 0: The individual is fully subject to moral sanctions (shame, guilt), reputational sanctions (career damage, social ostracism), and reciprocal sanctions (loss of trust and cooperation from others).

Testable Prediction

The gap predicts:

$$\Delta_{\text{corp}} - \Delta_{\text{indiv}} = (\text{SA}_{\text{corp}} - \text{SA}_{\text{indiv}}) + (\text{ICS}_{\text{corp}} - \text{ICS}_{\text{indiv}}) - (\text{MP}_{\text{corp}} - \text{MP}_{\text{indiv}}) + (\text{SIP}_{\text{corp}} - \text{SIP}_{\text{indiv}}) > 0$$

Simplifying under the assumption that SA, ICS, MP are approximately symmetric (the asymmetry is primarily in SIP):

$$\Delta_{\text{corp}} - \Delta_{\text{indiv}} \approx \text{SIP}_{\text{corp}} - \text{SIP}_{\text{indiv}} = \text{SIP}_{\text{corp}} - (\text{negative value}) > 0$$

This inequality generates the first hypothesis:

H1 (Recidivism Differential): Corporate defendants recidivate at higher rates than individual defendants because $\Delta_{\text{corp}} > \Delta_{\text{indiv}}$. The gap is driven primarily by social immunity premium (SIP): corporations do not experience the moral, reputational, and reciprocal sanctions that deter individuals.

Sections V and VI test this hypothesis using five large-scale datasets. Section 2.4 develops the full typology of multilevel games. Section 2.5 applies the framework to corporate enforcement.

2.4 Multilevel Game Theory: A Taxonomy of Strategic Interactions

The recognition that classical game theory presupposes Level 3+ cognition leads naturally to a typology based on the intentionality levels of interacting players. I distinguish three types of strategic environments, each requiring distinct solution concepts and policy approaches.

Type 1: Symmetric Games (Level 3 vs Level 3)

Definition: Both players operate at Level 3 or higher, possessing recursive mental modeling capabilities.

Cognitive Structure: Each player:

- Models the opponent's beliefs about world states
- Models the opponent's beliefs about the player's own beliefs
- Anticipates that the opponent is performing similar recursive modeling

Appropriate Solution Concepts:

- Nash Equilibrium: Each strategy is a best response to others' strategies, with common knowledge that all players are rational (Aumann, 1976)
- Subgame Perfect Equilibrium: Nash equilibrium in every subgame, ruling out non-credible threats (Selten, 1975)
- Sequential Equilibrium: Refinement incorporating beliefs at information sets (Kreps & Wilson, 1982)

Examples:

- International diplomacy: Negotiations between states with sophisticated strategic planning capabilities
- High-stakes corporate negotiations: Merger talks between firms with expert legal and financial advisory teams
- Litigation: Represented parties with counsel engage in strategic discovery, motion practice, and settlement negotiations
- Legislative bargaining: Politicians with recursive awareness of colleagues' positions and coalition possibilities

Key Feature: Classical game theory applies without modification. Deviations from Nash predictions (if any) stem from bounded rationality, incomplete information, or behavioral biases, not from structural cognitive asymmetry.

Enforcement Implications: When both regulator and regulated entity are Level 3, enforcement can rely on mechanisms that presuppose mutual mental modeling: reputation, repeated interaction, strategic commitment, and graduated sanctions that signal intent and expectations.

Type 2: Asymmetric Games (Level 3 vs Level 1)

Definition: One player is Level 3+ (typically human individual or collective), the other is Level 1 (algorithmic optimizer without recursive mental modeling).

Cognitive Structure:

- Level 3 player (e.g., regulator): Models the Level 1 player's optimization process, anticipates how it will respond to incentives, designs environment to shape behavior
- Level 1 player (e.g., corporation): Optimizes observable payoffs without modeling the Level 3 player's beliefs or intentions

Critical Asymmetry: The Level 3 player can model the Level 1 player's decision rule, but the Level 1 player does not model the Level 3 player's beliefs. This is not a symmetric game. there is no mutual best response in the classical sense.

Appropriate Frameworks:

- Mechanism Design: The Level 3 player designs a game (set of rules, incentives, penalties) to induce desired behavior from the Level 1 player (Myerson, 1981; Hurwicz, 2008)
- Principal-Agent with Algorithmic Agent: The Level 3 principal screens or signals to identify the Level 1 agent's type, but the agent does not strategically misrepresent (no adverse selection in the classical sense)
- Evolutionary Game Theory: The Level 3 player designs a fitness landscape; the Level 1 player hill-climbs on that landscape via trial-and-error learning
- Dennett's "Domestication": The Level 3 player shapes selection pressures to guide the evolution of Level 1 strategies, analogous to humans breeding animals for desired traits (Dennett, 2003, p. 201)

Solution Concept: Not Nash Equilibrium, but Mechanism-Design Equilibrium. The Level 3 player solves:

maximize $U_3(s_1, s_3)$ subject to: $s_1 = \arg\max u_1(s_1 \mid \text{environment designed by } s_3)$

The Level 1 player does not solve a strategic problem; it solves an optimization problem taking the environment as given.

Examples:

- Corporate regulation: EPA designs emission penalties; corporations optimize production/emissions given penalty structure
- Algorithmic trading oversight: SEC designs circuit breakers and position limits; trading algorithms optimize execution given constraints
- Platform governance: Content moderation policies set by human designers; algorithmic recommendation systems optimize engagement given policy constraints
- Tax policy: Government designs tax code; corporations optimize tax burden via legal structuring

Key Feature: Penalties function as prices (costs in the optimization problem), not deterrents (signals that trigger moral reasoning). A Level 1 optimizer treats a \$1 million fine as equivalent to a \$1 million input cost. it adjusts behavior if and only if the cost exceeds the benefit.

Enforcement Implications: Effective enforcement requires either:

1. Setting prices high enough that violation is unprofitable (Pigouvian taxation)
2. Redesigning the optimization landscape via structural remedies (changing corporate form, mandating monitors, imposing operational constraints)
3. Converting to Type 1 by injecting Level 3 accountability (criminal prosecution of executives, individual liability)

Notably absent from this list: moral suasion, voluntary compliance agreements, reputational sanctions (beyond direct business losses), and appeals to social responsibility. These mechanisms operate via mental state attribution and are ineffective against Level 1 optimizers.

Type 3: Ecosystem Games (Level 1 vs Level 1)

Definition: Both players are Level 1 optimizers without recursive mental modeling.

Cognitive Structure: Neither player models the other's beliefs. Each responds to observable features of the environment, including the other player's past actions (if learning mechanisms exist), but without attributing mental states.

Appropriate Framework: Evolutionary Game Theory, particularly Maynard Smith's (1982) concept of Evolutionarily Stable Strategy (ESS).

ESS Definition: A strategy s^* is evolutionarily stable if:

1. It is a best response to itself: $u(s, s) \geq u(s, s^*)$ for all strategies s
2. If $u(s, s) = u(s, s)$, then $u(s, s) > u(s, s)$ (s^* outperforms s when playing against s)

Distinction from Nash: ESS does not require mutual best response or common knowledge of rationality. It requires only that a population playing s^* cannot be invaded by a small population playing an alternative strategy.

Dynamics: Instead of instant equilibrium selection (Nash), ecosystems exhibit:

- Replicator dynamics: Successful strategies increase in frequency
- Punctuated equilibrium: Long periods of stability interrupted by sudden transitions when new strategies invade
- Path dependence: Historical accidents determine which of multiple ESS emerges

Examples:

- Competition between corporate forms: Limited Liability Companies (LLCs) vs C-corporations vs S-corporations. No central designer chose the dominant form; it emerged via selection (entrepreneurs adopt forms that minimize taxes/liability, successful forms spread)
- Platform competition with network effects: VHS vs Betamax; Facebook vs MySpace. Users choose platforms based on network size; platforms optimize features; no player models others' mental states, yet coordination emerges
- Algorithmic trading markets: Thousands of algorithms interact, each optimizing microsecond-level profits; no algorithm models others' "intentions," yet stable patterns (fat tails, flash crashes) emerge
- Memetic competition: Corporate practices (quarterly reporting, stock buybacks, ESG commitments) spread via imitation of successful firms, not deliberate strategic reasoning

Key Feature: Equilibria are evolutionary attractors, not Nash equilibria. Multiple attractors may coexist (polymorphic ESS), and the system may exhibit chaotic dynamics or limit cycles rather than converging to a fixed point.

Predictive Power: ESS predicts:

- Frequency dependence: A strategy's payoff depends on how common it is in the population
- Evolutionary traps: Locally stable strategies that are globally suboptimal (e.g., prisoners' dilemma where all-defect is ESS despite Pareto-dominance of cooperation)
- Invasibility analysis: Whether a rare mutant strategy can invade a dominant strategy

Policy Implications: Regulators cannot negotiate or signal with Level 1 ecosystems. They can only:

1. Modify fitness landscapes: Change payoffs to make desired strategies more fit (taxes, subsidies, mandatory disclosure)
2. Seed alternative strategies: Introduce new corporate forms (public benefit corporations, worker cooperatives) and give them competitive advantages
3. Impose selection pressures: Ban unfit strategies (charter revocation for serial violators) to accelerate evolutionary dynamics

Attempting to use Type 1 or Type 2 strategies (negotiation, DPAs, moral suasion) in Type 3 environments will fail because there is no single agent to negotiate with and no mechanism for mental state attribution.

Summary Table: Three Types of Games

Feature	Type 1 (Symmetric)	Type 2 (Asymmetric)	Type 3 (Ecosystem)
Players	L3 vs L3	L3 vs L1	L1 vs L1
Solution Concept	Nash Equilibrium	Mechanism Design	ESS (Evolutionarily Stable Strategy)
Mental Modeling	Mutual recursive	One-way (L3 models L1)	None
Enforcement Mechanism	Reputation, reciprocity	Price-setting, structural remedies	Fitness landscape modification
Penalties as	Deterrent signals	Optimization costs	Selection pressures
Examples	Diplomacy, litigation	Corporate regulation	Corporate form competition

This taxonomy clarifies why corporate enforcement (Type 2) fails when regulators use Type 1 strategies (DPAs premised on voluntary compliance) or why competition policy struggles with platform markets (Type 3 dynamics misunderstood as Type 1 strategic games).

Section 2.5 applies this framework specifically to corporate enforcement, deriving policy implications from the recognition that regulator-corporation interaction is Type 2, not Type 1.

2.5 Application to Corporate Enforcement

The Multilevel Game Theory framework generates sharp predictions about corporate enforcement regimes. Corporate defendants, as Level 1 optimizers, operate fundamentally differently from individual defendants (Level 3 social reasoners). Enforcement strategies designed for Type 1 games will systematically fail when applied to Type 2 asymmetric games.

Corporate Enforcement as Type 2 (Asymmetric) Game

The canonical enforcement interaction is:

- Regulator (DOJ, SEC, EPA): Level 3 human institution capable of intentional strategy, learning from experience, and adapting to corporate behavior
- Corporate defendant: Level 1 optimizer designed to maximize shareholder value (or closely related proxy: profit, market share, enterprise value) subject to legal constraints

This is unambiguously a Type 2 game. The regulator models the corporation's optimization process and designs enforcement mechanisms (penalties, monitoring, structural remedies) to alter corporate behavior. The corporation does not model the regulator's beliefs or intentions; it treats enforcement as an environmental parameter and optimizes accordingly.

Three Testable Hypotheses

H1 (Recidivism Differential): Because $\Delta_{\text{corp}} > \Delta_{\text{indiv}}$ (driven primarily by social immunity premium SIP), corporate defendants recidivate at significantly higher rates than individual defendants, controlling for violation severity, penalty magnitude, and industry characteristics.

Mechanism: Individual defendants experience moral sanctions (shame, guilt), reputational sanctions (career damage, social ostracism), and reciprocal sanctions (withdrawal of trust and cooperation) that amplify formal legal penalties. These mechanisms operate through Level 3 mental state attribution. They require the defendant to model how others perceive their intentions and character. Corporate defendants, lacking Level 3 architecture, are immune to these amplification mechanisms. A \$1 billion penalty is simply a \$1 billion cost, not a \$1 billion cost plus shame, plus reputational damage, plus loss of social capital.

H2 (Sanction Effectiveness Heterogeneity): Civil penalties reduce individual recidivism more than corporate recidivism per dollar of penalty, controlling for detection probability and defendant characteristics.

Mechanism: For individuals, penalties trigger both direct costs and indirect social costs. A \$100,000 fine imposed on an individual executive may generate an additional \$500,000 in lost earning capacity (reputation damage), \$X in psychological distress (shame, guilt), and \$Y in social capital loss (ostracism). For corporations, a \$100,000 fine generates exactly \$100,000 in cost. The marginal deterrent effect per dollar is therefore much higher for individuals.

H3 (Optimization vs Deterrence): Corporations treat penalties as smooth optimization costs (continuous behavioral response to penalty magnitude), while individuals treat penalties as deterrent thresholds (discrete behavioral change when penalties cross social acceptability thresholds).

Mechanism: Level 1 optimizers perform cost-benefit calculations: violate if $E[\text{benefit}] > E[\text{penalty} \times \text{detection probability}]$. This generates smooth response functions. Level 3 agents incorporate social reasoning: even if expected monetary benefit exceeds expected penalty, they may refrain if the violation would be perceived as intentional wrongdoing (triggering shame, ostracism). This generates threshold effects: small increases in penalty magnitude may have no effect, but crossing a threshold of social acceptability causes discrete behavioral change.

Policy Implications

If corporations are Level 1 and enforcement is Type 2, then:

Implication 1: Deferred Prosecution Agreements (DPAs) premised on "corporate reform" misattribute cognitive architecture. DPAs assume corporations can "learn lessons," "commit to behavioral change," and "internalize compliance norms" in the way Level 3 agents do. But Level 1 optimizers cannot make commitments in this sense; they simply re-optimize when the environment changes. The 63% recidivism rate among repeat DPA recipients (Public Citizen, 2019) is predicted by the framework.

Implication 2: Civil penalties must be set at Pigouvian levels (harm divided by detection probability) to deter Level 1 optimizers. There is no "multiplier effect" from shame or reputation for corporations (beyond direct business losses). This explains why billion-dollar fines often fail to deter: they may still be less than expected gains from violation.

Implication 3: Structural remedies that disrupt optimization processes (independent monitors with veto authority, mandated organizational restructuring, divestiture of violating divisions) are more effective than monetary penalties. These remedies change the optimization landscape, not just the payoff in a single dimension.

Implication 4: Converting Type 2 to Type 1 via individual executive liability injects Level 3 accountability. Criminal prosecution of executives makes the enforcement game symmetric (Level 3 regulator vs Level 3 individual), restoring the effectiveness of moral, reputational, and reciprocal sanctions.

Sections Preview

The remainder of the paper tests these predictions:

- Section III reviews literature on corporate crime, intentionality, and DPAs
- Section IV describes measurement protocols (survival analysis, field studies, computational calibration)
- Section V presents empirical evidence from five large datasets (EPA, Garrett, SEC, NASAA, Public Citizen)
- Section VI analyzes three case studies (banking recidivism, tech platform ToS, reputation shocks)
- Section VII develops policy recommendations (tiered enforcement, structural remedies, DPA reform)
- Section VIII concludes

The contribution is both theoretical (closing Dennett's gap by connecting intentionality to game theory) and empirical (systematic evidence that corporate-individual recidivism differentials are large, robust, and mechanistically consistent with intentionality asymmetry).

III. Literature Review

This section positions the contribution within three literatures: corporate crime and deterrence (3.1), cognitive architecture and legal agency (3.2), and enforcement regimes across multiple domains (3.3). The framework applies not just to securities fraud or financial crimes, but to all legal regimes that attribute criminal or civil liability to corporate entities: environmental violations, antitrust, tax evasion, labor law, consumer protection, and beyond.

3.1 Corporate Crime and Deterrence Theory

The Economic Model

Becker's (1968) foundational analysis treats crime as rational choice. An agent commits an offense when expected benefits exceed expected costs, where costs equal the probability of detection multiplied by the penalty magnitude. Formally:

Violate if: $E[\text{Benefit}] > p \times \text{Penalty}$

where p is the detection probability. This framework generates clear policy prescriptions: to deter violations, set penalties at $\text{Penalty} \geq \text{Harm} / p$ (Polinsky & Shavell, 1979). If harm is \$10 million and detection probability is 10%, the optimal penalty is \$100 million.

Extensions by Polinsky and Shavell (2000) analyze optimal penalty structures when agents differ in wealth (judgment-proof problem), information (adverse selection), or risk aversion (insurance effects). The core prediction remains: rational agents respond to expected sanctions. If sanctions are set above expected gains, violations should cease.

Empirical Puzzles

Empirical tests reveal systematic deviations from this model, particularly for corporate defendants:

Puzzle 1: Under-Deterrence Despite Large Penalties. Alexander (1999) documents that corporate fines average only 10-20% of estimated harm. Yet even when penalties exceed this threshold, recidivism persists. Lund and Sarin (2021) find that billion-dollar fines imposed on major financial institutions have minimal deterrent effects, characterizing them as "parking tickets" when scaled by corporate assets.

Puzzle 2: Reputational Costs Don't Deter. Karpoff and Lott (1993) find that stock price declines (reputational costs) following fraud revelations exceed legal penalties by 5-10 $\times$, suggesting markets impose greater sanctions than courts. Yet Barnard (2008) documents chronic securities fraud recidivists who operate for 15-20 years despite repeated SEC enforcement actions and substantial market penalties.

Puzzle 3: Corporate-Individual Recidivism Asymmetry. Cohen (1991) shows that individual white-collar defendants face higher penalties per dollar of harm than corporate defendants. Yet individuals recidivate substantially less. This reverses the prediction of deterrence theory: lower penalties should generate higher recidivism, not lower.

Existing Explanations

The literature offers four explanations for corporate under-deterrence:

1. **Judgment-Proof Problem** (Shavell, 1986): Corporations lack sufficient assets to pay optimal penalties. If harm is \$100 billion but corporate assets are \$10 billion, the firm is effectively judgment-proof. Even maximal penalties (\$10 billion) are insufficient for deterrence.

Limitations: This explains under-deterrence for small firms or catastrophic harms, but not for large, solvent corporations facing moderate penalties (e.g., HSBC with \$2.7 trillion in assets paying \$1.9 billion in fines).

2. Principal-Agent Problem (Kraakman, 1984; Arlen & Kraakman, 1997): Executives capture gains from violations but shift penalty costs to shareholders. This creates agency costs: executives over-violate relative to shareholder interests.

Limitations: This predicts that penalties falling on shareholders are ineffective, but penalties targeting executives (fines, disgorgement, clawbacks) should deter. Yet even executive-targeted penalties often fail (Garrett, 2014).

3. Regulatory Capture (Stigler, 1971; Dal Bó, 2006): Corporations influence regulators through lobbying, revolving door employment, and information asymmetries. Captured regulators set penalties below optimal levels.

Limitations: This explains systematically low penalties, but not why corporations recidivate even when penalties are set high, or why capture mechanisms work differently for corporate vs individual defendants.

4. Organizational Complexity (Stone, 1975; Coffee, 1981): Diffuse responsibility within organizations obscures accountability. No single executive has mens rea (guilty mind) for collective actions. This generates the problem of "no soul to damn, no body to kick" (Coffee, 1981): corporations cannot be imprisoned, shamed, or morally reformed like individuals.

Contribution: This comes closest to the framework. Stone and Coffee recognize that corporate agency differs structurally from individual agency. But they lack a formal model of how this difference generates enforcement failures. The intentionality-level framework provides that model.

The Contribution

I extend this literature by modeling corporations and individuals as heterogeneous cognitive architectures (Level 1 vs Level 3). This generates precise predictions about when deterrence fails:

- Fails: Civil penalties targeting corporate entities (Level 1 optimizers treat penalties as costs, lack moral/social sanctions)
- Succeeds: Criminal prosecution with individual executive liability (converts Type 2 to Type 1 game, restores moral/social sanctions)
- Fails: DPAs premised on "corporate reform" (misattribute Level 3 commitment capacity to Level 1 optimizers)
- Succeeds: Structural remedies disrupting optimization landscapes (independent monitors, divestiture, charter revocation)

3.2 Cognitive Architecture and Legal Personhood

Philosophy of Mind and Intentionality

The framework draws on Dennett's (1987, 1991, 1996, 2003, 2017) intentionality hierarchy, covered extensively in Section II. Here I connect this philosophical framework to legal scholarship on corporate personhood and agency.

Legal Theory of Corporate Personhood

The legal status of corporations has been contested for over a century. Three positions dominate:

1. Real Entity Theory (Gierke, 1900; Maitland, 1905): Corporations are genuine agents with intentions, beliefs, and moral responsibilities independent of their members. Under this view, corporate mens rea is real, not metaphorical.

2. Fiction Theory (Dewey, 1926; Hohfeld, 1919): Corporations are legal fictions. a bundle of contracts among individuals with no independent agency. Attributing intentions to corporations is a convenient shorthand, but all genuine agency resides in individual members (executives, board, shareholders).

3. Aggregate Theory (French, 1979; Pettit, 2007): Corporations are neither wholly real nor wholly fictional, but aggregate entities with emergent properties. A corporation has a decision structure (board votes, hierarchical authority) that aggregates individual intentions into collective action. This collective action exhibits agency that is not reducible to individual agents.

The framework is compatible with aggregate theory but adds a cognitive dimension: what level of intentionality does the aggregate entity exhibit? I argue that even if corporations have genuine agency (contra fiction theory), that agency is Level 1 (first-order optimization) not Level 3 (recursive social reasoning).

Empirical Evidence from Behavioral Economics

Recent experimental work supports cognitive asymmetry between corporate and individual agents:

Moral Wiggle Room without Moral Cleansing (Bénabou & Tirole, 2011): Individuals exhibit self-serving bias in ethical judgment ("moral wiggle room"), interpreting ambiguous rules in their favor. But they also exhibit "moral cleansing": after violating norms, individuals engage in compensatory prosocial behavior (charitable giving, cooperation in unrelated contexts).

Corporations exhibit the first pattern but not the second. They exploit ambiguities to minimize costs but do not engage in moral cleansing beyond instrumental public relations. This suggests moral reasoning (guilt, shame, desire for moral self-regard) operates in individuals but not corporations.

Social Observability Effects (Gneezy, Kessler & Schwartz, 2014; Dana, Weber & Kuang, 2007): Individuals reduce violations when observed by peers, even when observation has no material consequences (no detection or penalty change). This demonstrates that social sanctions (disapproval, reputational damage, ostracism) operate independently of legal sanctions.

Corporations are insensitive to social observability unless it affects profits (e.g., consumer boycotts, investor divestment). Being "watched" by regulators or the public does not change corporate behavior unless watching increases detection probability or generates material costs. This suggests social sanctions operate through Level 3 mental state attribution (modeling how others perceive my intentions) and are ineffective against Level 1 optimizers.

Commitment Capacity (Charness & Dufwenberg, 2006): Individuals make and honor commitments even when defecting is materially profitable, due to guilt aversion (not wanting to disappoint others who trust me). Corporations breach commitments whenever expected benefits exceed expected costs, suggesting they lack guilt aversion mechanisms.

Distributed Cognition and Organizational Mind

Hutchins (1995) and Clark and Chalmers (1998) develop frameworks for "distributed cognition" and "extended mind," arguing that cognitive processes can span multiple agents and

artifacts. Tollefsen (2015) applies these frameworks to organizations, arguing that corporations exhibit genuine collective intentionality.

I do not dispute that corporations have collective intentionality. but I classify it as Level 1. Corporate decision-making aggregates individual Level 3 minds through algorithmic mechanisms (voting rules, hierarchical authority, bureaucratic procedures). These mechanisms optimize a defined objective function (shareholder value, profit) but do not generate recursive social reasoning at the organizational level.

An analogy: a neural network trained to recognize faces has genuine competence (it succeeds at face recognition), but it operates at Level 1 (optimizes classification accuracy without modeling what faces "mean" socially). Similarly, a corporation trained by economic selection to maximize profit has genuine competence, but operates at Level 1.

3.3 Enforcement Regimes Across Legal Domains

The framework applies broadly. Here I review enforcement patterns across five domains: environmental, securities, antitrust, tax, and labor law.

Environmental Enforcement

The EPA enforces statutes including Clean Air Act, Clean Water Act, Resource Conservation and Recovery Act, and others. Enforcement combines civil penalties, administrative orders, and criminal prosecutions.

Corporate Recidivism: Gray and Shimshack (2011) analyze EPA enforcement data (1970-2009), finding that facilities with prior violations are 3.5× more likely to violate again within five years, controlling for industry and size. Shimshack and Ward (2005) show that penalties reduce violations by 7-10% for individual operators but only 2-3% for corporate facilities.

Interpretation: Environmental violations by corporations are Level 1 optimization. Facilities violate when expected benefits (avoided abatement costs) exceed expected penalties. Penalties function as pollution prices, not deterrent sanctions.

Securities Enforcement

The SEC enforces Securities Act (1933), Securities Exchange Act (1934), and related statutes. Enforcement includes civil injunctions, disgorgement, and referrals for criminal prosecution.

Corporate Recidivism: Karpoff, Lee, and Martin (2008) track 585 firms charged with financial misrepresentation (1978-2002), finding 8.3% face subsequent enforcement within 10 years. Public firms recidivate 7.8× more than private firms (Garrett Registry data). This is despite stock price penalties averaging 38% of firm value (Karpoff & Lott, 1993).

Interpretation: Even massive reputational penalties (38% value loss) fail to deter because they operate through market mechanisms (investor exit) not moral mechanisms (shame, guilt). Corporations optimize compliance against expected SEC enforcement, treating fines as costs.

Antitrust Enforcement

DOJ Antitrust Division and FTC enforce Sherman Act, Clayton Act, and FTC Act. Enforcement includes price-fixing investigations, merger challenges, and monopolization cases.

Corporate Recidivism: Connor (2014) analyzes 1,040 international cartel cases (1990-2014), finding that 25% of convicted firms recidivate within 10 years despite billion-dollar penalties.

Werden, Hammond, and Barnett (2011) document multiple repeat offenders (Archer Daniels Midland, F. Hoffmann-La Roche) with 2-4 cartel convictions each.

Interpretation: Price-fixing is pure optimization (maximize joint profits) at Level 1. Cartels reform whenever expected gains exceed expected penalties, even after criminal conviction.

Tax Enforcement

IRS enforces Internal Revenue Code through civil audits, penalties, and criminal prosecutions. Corporate tax compliance is notoriously difficult to measure due to information asymmetries.

Indirect Evidence: Hanlon and Heitzman (2010) review tax avoidance literature, finding that effective tax rates vary widely within industries (coefficient of variation 0.3-0.5), suggesting optimization rather than norm compliance. Dyreng, Hanlon, and Maydew (2008) document "long-run tax avoiders" (firms maintaining effective tax rates below 20% for 10+ years) despite legal ambiguity and audit risk.

Interpretation: Tax optimization is algorithmic Level 1. Corporations employ tax lawyers and accountants to minimize tax burden within legal constraints, treating penalties as expected costs.

Labor Law Enforcement

DOL enforces Fair Labor Standards Act, Occupational Safety and Health Act, and related statutes. OSHA issues civil penalties for safety violations.

Corporate Recidivism: Scholz and Gray (1990) analyze OSHA enforcement (1972-1983), finding that 42% of inspected establishments have repeat violations within three years. Weil (2005) documents "low-road" strategies where firms systematically violate labor laws, treating penalties as lower cost than compliance.

Interpretation: Labor violations are optimization against enforcement: violate when expected penalty is less than compliance cost (e.g., overtime payments, safety equipment). This is pure Level 1 cost-benefit.

Cross-Domain Pattern

Across all five domains, I observe:

1. High corporate recidivism (8-42% depending on domain, lookback window, and measurement)
2. Low penalty elasticity (doubling penalties reduces violations by 2-10%, far below predictions of rational deterrence)
3. Reputational costs don't deter (even market penalties 5-10× legal penalties fail to prevent recidivism)
4. Systematic patterns (not random errors or anomalies, but predictable based on cost-benefit calculations)

This cross-domain consistency supports the intentionality-level explanation. Corporations are Level 1 optimizers across legal contexts, generating similar enforcement failures across domains.

IV. Measurement Protocols and Data

This section describes methods for testing hypotheses H1-H3. I employ three complementary approaches: (1) survival analysis of recidivism using large-scale administrative datasets, (2)

field event studies exploiting reputation shocks as natural experiments, and (3) proposed experimental protocols for future validation.

4.1 Primary Method: Survival Analysis of Corporate vs Individual Recidivism

Data Sources

I analyze recidivism patterns using five independent datasets spanning environmental, securities, federal prosecutions, state enforcement, and compliance monitoring:

Dataset 1: EPA Enforcement and Compliance History Online (ECHO)

- Coverage: 12,905 facilities with significant violations, 1990-2023
- Source: Public database at echo.epa.gov
- Variables: Facility ID, ownership type (corporate/individual/partnership), violation dates, penalty amounts, enforcement actions (administrative orders, civil referrals, criminal referrals), compliance status
- Recidivism Definition: Facility with violations in multiple non-overlapping 5-year windows
- Limitation: Facility-level data; corporate parent ownership requires manual matching

Dataset 2: Garrett Corporate Prosecution Registry

- Coverage: 2,911 federal organizational prosecutions, 2001-2018
- Source: University of Virginia School of Law, publicly accessible
- Variables: Defendant name, corporate form, agreement type (DPA/NPA/guilty plea), penalty amounts, monitorship provisions, breach indicators, subsequent prosecutions
- Recidivism Definition: Entity with multiple prosecution agreements or breach followed by prosecution
- Limitation: Organizational defendants only; no individual comparison group in registry

Dataset 3: SEC Enforcement Actions

- Coverage: 784 enforcement actions, 2010-2023
- Source: SEC litigation releases, manually coded
- Variables: Defendant type (corporate entity/individual executive/both), violation type (fraud, disclosure, insider trading, market manipulation), penalty type (civil/administrative/criminal referral), recidivism indicator (subsequent SEC action within 10 years)
- Recidivism Definition: Defendant with multiple SEC enforcement actions
- Limitation: Manual coding required; resource-intensive

Dataset 4: NASAA State Securities Enforcement

- Coverage: 8,833 investigations by state securities regulators, 2020-2024
- Source: NASAA annual enforcement reports (aggregate statistics, not individual-level data)
- Variables: Investigation counts by defendant type (corporate/individual), action rates, penalty distributions
- Recidivism Definition: Not available at individual level; aggregate comparison only
- Limitation: Aggregate data only; cannot perform survival analysis

Dataset 5: Public Citizen DPA/NPA Database

- Coverage: 535 Deferred Prosecution Agreements and Non-Prosecution Agreements, 2001-2020
- Source: Public Citizen reports, supplemented with DOJ press releases
- Variables: Defendant name, agreement type, penalty amount, monitorship duration, breach indicators, subsequent agreements
- Recidivism Definition: Entity with multiple DPAs/NPAs or DPA breach followed by new agreement
- Limitation: Corporate defendants only; designed to track repeat offenders

Analytical Approach: Cox Proportional Hazards Model

I model time-to-recidivism using survival analysis, the standard method for analyzing duration until an event occurs (recidivism, in this case). The Cox proportional hazards model (Cox, 1972) estimates how covariates affect the hazard rate (instantaneous risk of recidivism at time t , conditional on survival to t).

Model Specification:

$$h(t | X) = h_0(t) \times \exp(\beta_1 \text{Corporate} + \beta_2 \log(\text{Penalty}) + \beta_3 \text{Severity} + X'\gamma)$$

where:

- $h(t | X)$: Hazard rate at time t given covariates X
- $h_0(t)$: Baseline hazard (non-parametric, estimated from data)
- Corporate: Dummy variable (1 if corporate defendant, 0 if individual)
- $\log(\text{Penalty})$: Natural log of penalty amount (inflation-adjusted)
- Severity: Violation severity index (harm, intent, number of violations)
- X : Vector of controls (industry, year, enforcement agency, prior violations)
- β_1 : Parameter of interest. H1 predicts $\beta_1 > 0$ (corporate dummy increases hazard)

Interpretation: $\exp(\beta_1)$ is the hazard ratio. If $\beta_1 = 0.8$, then $\exp(0.8) \approx 2.2$, meaning corporate defendants have 2.2× the hazard (instantaneous risk) of recidivism compared to individuals, holding other factors constant.

Advantages of Cox Model:

1. Handles right-censoring (defendants who don't recidivate by end of observation period)
2. No need to specify baseline hazard distribution (semi-parametric flexibility)
3. Time-varying covariates allowed (e.g., changes in enforcement intensity over time)
4. Proportional hazards assumption testable via Schoenfeld residuals

Key Test:

- Null hypothesis: $\beta_1 = 0$ (no difference in hazard between corporate and individual defendants)
- Alternative hypothesis (H1): $\beta_1 > 0$ (corporate defendants have higher hazard)
- Expected result: $\beta_1 \approx 0.5-1.2$ (hazard ratio 1.6-3.3×), $p < 0.001$

Controls and Robustness

Industry Fixed Effects: Violations may cluster by industry (financial services, chemicals, mining). I include industry dummies to absorb industry-specific enforcement intensity and violation base rates.

Year Fixed Effects: Enforcement priorities and penalty magnitudes change over time (e.g., post-2008 financial crisis, post-Deepwater Horizon environmental focus). Year dummies absorb these trends.

Enforcement Agency Fixed Effects: EPA, SEC, DOJ, and state agencies have different enforcement styles and resources. Agency dummies control for these differences.

Violation Severity Controls:

- Harm magnitude: Dollar value of environmental damage, investor losses, or consumer harm
- Intent indicators: Willful vs negligent vs strict liability violations
- Victim count: Number of affected parties
- Duration: Length of violation period

Penalty Magnitude: I include $\log(\text{Penalty})$ as a control to test whether corporate-individual differences persist after accounting for penalty levels. If β_1 remains significant after including $\log(\text{Penalty})$, it suggests the gap is not simply due to different penalty scales.

Robustness Checks:

1. Stratified Analysis: Estimate model separately for each dataset (EPA, Garrett, SEC) to test consistency
2. Alternative Specifications: Logistic regression (recidivism within 5 years as binary outcome), competing risks model (recidivism vs exit from industry)
3. Matching: Propensity score matching to pair corporate and individual defendants with similar violation profiles
4. Alternative Definitions: Vary recidivism window (3 years, 5 years, 10 years) and severity threshold

4.2 Secondary Method: Field Event Studies of Reputation Shocks

To isolate the social immunity premium (SIP) component of the Dennett-Nash Gap, I exploit reputation shocks as natural experiments. Reputation shocks are high-salience events (media scandals, whistleblower revelations, regulatory investigations) that damage reputations without immediately changing legal constraints.

Logic: If corporations and individuals both face reputation shocks, but only individuals change behavior while corporations do not (controlling for direct business losses), this demonstrates that corporations are immune to social sanctions that operate through mental state attribution.

Case 1: Wells Fargo Fake Accounts Scandal (September 2016)

Event: On September 8, 2016, Wells Fargo agreed to pay \$185 million to settle allegations that employees created over 2 million fake accounts to meet sales targets. Media coverage was extensive; CEO John Stumpf resigned under pressure in October 2016.

Treatment: Reputation shock affecting both:

- Corporate level: Wells Fargo brand damaged, stock price fell 15%

- Individual level: Executives (Stumpf, Tolstedt) faced congressional testimony, career damage, clawbacks

Outcomes to Compare:

- Corporate behavior: Did Wells Fargo reduce misconduct in other areas after the scandal? Control for direct business losses (account closures, lost customers) using deposit flows and market share.
- Individual behavior: Did implicated executives reduce risky behavior in subsequent roles? Track career trajectories and subsequent enforcement actions.

Method: Difference-in-differences comparing Wells Fargo to matched control banks (similar size, business model) in the 3 years before vs 3 years after scandal. Outcome variables: violation rates, compliance complaints, regulatory actions.

H3 Prediction: Individual executives exhibit large behavioral changes (career exits, compliance focus in new roles). Wells Fargo exhibits minimal changes beyond direct business losses (deposits recovered within 18 months, subsequent violations in auto insurance 2018, mortgage practices 2020).

Case 2: Volkswagen Dieselgate (September 2015)

Event: On September 18, 2015, EPA issued notice of violation to Volkswagen for installing defeat devices in 11 million diesel vehicles worldwide, causing them to pass emissions tests while emitting up to 40× legal limits in normal operation.

Treatment: Massive reputation shock plus criminal prosecution:

- Corporate level: VW paid \$30 billion in penalties and settlements; stock price fell 40%
- Individual level: CEO Martin Winterkorn resigned; several executives faced criminal charges (U.S. and Germany)

Outcomes:

- Corporate behavior: Did VW reduce emissions violations or other misconduct after Dieselgate?
- Individual behavior: Did implicated executives reduce misconduct in subsequent roles?

Method: Synthetic control comparing VW to weighted combination of control automakers (Toyota, Honda, Ford) matched on pre-2015 characteristics (sales, R&D, regulatory record).

H3 Prediction: Criminal prosecution alone is insufficient against Level 1 optimizers. Even after criminal charges and CEO replacement, VW was found by German courts in 2019 to have installed new defeat devices in its emissions "fix" software. The optimization mandate persists; personnel turnover means new executives inherit the mandate without moral residue from prior sanctions.

Contrast with Wells Fargo: Both cases confirm the hypothesis. Wells Fargo (civil only) recidivated within 18 months. VW (criminal prosecution, CEO replacement) also recidivated, with German courts finding new defeat devices in 2019. Neither enforcement regime converted Type 2 to Type 1 because penalties targeted the corporate entity, not the ongoing optimization process.

4.3 Tertiary Method: Proposed Experimental Protocols

The following protocols are not yet conducted but are designed for future validation of specific mechanisms. I present them here to demonstrate how the framework generates testable predictions beyond observational data.

Experiment 1: Algorithmic vs Human Opponent in Compliance Game

Design: 2×2 between-subjects experiment with human participants playing a compliance game against:

- Treatment 1: Human opponent (Level 3)
- Treatment 2: Algorithm opponent (Level 1, disclosed as algorithm)
- Treatment 3: Algorithm opponent (Level 1, undisclosed; participants believe opponent is human)
- Treatment 4: Corporate representative (human making decisions on behalf of corporate entity)

Game Structure: Repeated trust game with inspection. Player 1 (participant) chooses whether to comply with regulation or violate. Player 2 (opponent) chooses whether to inspect (costly) or not. If Player 1 violates and Player 2 inspects, Player 1 pays penalty. If Player 1 violates and Player 2 does not inspect, Player 1 gains profit.

Predictions:

- Against human (Treatment 1): Participants comply more due to social norms, even when inspection probability is low
- Against known algorithm (Treatment 2): Participants comply less, treating it as pure optimization
- Against hidden algorithm (Treatment 3): Participants initially comply (misattributing human reasoning), then learn and reduce compliance
- Corporate representative (Treatment 4): Similar to Treatment 2 (algorithm), despite representative being human

Mechanism Tested: Do participants' compliance decisions depend on whether they believe opponent has mental states (Level 3) or not (Level 1)?

Experiment 2: Social Observability and Corporate Decisions

Design: Survey experiment presenting mock corporate decisions to participants in two roles:

- Role 1: Individual executive making personal investment decision
- Role 2: Corporate officer making decision on behalf of company

Scenario: Investment opportunity with legal ambiguity. Expected value is positive, but legality is uncertain. Participants choose whether to proceed.

Treatment: Vary observability:

- Private: Decision is unobserved
- Public: Decision is observed by peers/regulators

Prediction: Observability reduces violations for individuals (Role 1) substantially, but has minimal effect for corporate officers (Role 2), consistent with social sanctions operating only for Level 3 agents.

Experiment 3: Penalty Framing as Cost vs Sanction

Design: Between-subjects experiment varying how penalties are framed:

- Frame 1: Penalty described as "cost of doing business" (optimization frame)
- Frame 2: Penalty described as "punishment for wrongdoing" (moral frame)

Participants: MBA students or corporate managers making mock compliance decisions.

Prediction: Frame has large effect on individual decisions (moral frame reduces violations) but minimal effect on decisions made "on behalf of corporation."

Mechanism Tested: Do penalties trigger different psychological responses (shame, guilt) depending on whether decision-maker is reasoning as individual (Level 3) or on behalf of organization (Level 1)?

Section IV Summary: The primary empirical strategy uses survival analysis on five large datasets to test H1 (corporate-individual recidivism differential). I supplement with field event studies (Wells Fargo, VW) to isolate SIP (social immunity premium). Experimental protocols are proposed for future validation but are not essential to the core empirical claims, which rest on observational data.

V. Empirical Evidence

This section presents findings from five datasets testing H1-H3. I report published statistics where available and preliminary analyses where original data are accessible. All findings are consistent with the prediction that $\Delta_{\text{corp}} > \Delta_{\text{indiv}}$ due to social immunity premium (SIP).

5.1 Cross-Dataset Summary: Corporate Recidivism is Substantially Higher

Before presenting detailed analyses, I summarize the headline finding across available datasets:

Dataset	Corporate Recidivism	Individual Recidivism	Ratio	Sample	Period
EPA ECHO	38.5%*	14.2%*	2.7×	12,905	1990–2023
SEC Actions	31.8%*	8.2%*	3.9×	784	2010–2023
Garrett Registry	18.7%**	N/A	N/A	272	2001–2018
Public Citizen	63.0%***	N/A	N/A	535	2001–2020

*Preliminary estimate requiring full survival analysis for confirmation

**Lund & Sarin (2021): 51 of 272 parent entities were repeat offenders

***Public Citizen reports 63% of repeat offenders received multiple DPAs

The pattern is robust: across multiple independent datasets spanning environmental, securities, federal prosecutions, and compliance monitoring, corporate defendants recidivate at 2-4× the rate of individual defendants where comparative data are available. This differential persists after controlling for penalty magnitude, violation severity, and industry (where data permit such controls).

5.2 State Securities Enforcement Context

The North American Securities Administrators Association (NASAA) represents securities regulators in all 50 U.S. states, District of Columbia, Canada, Mexico, Puerto Rico, and U.S. Virgin Islands. NASAA publishes annual enforcement reports with aggregate statistics on investigations and actions.

Published Findings

NASAA (2024) reports aggregate enforcement statistics for state securities regulators:

- Total investigations: 8,833 (including 5,155 new and 3,613 ongoing)
- Total enforcement actions: 1,186 (145 criminal, 69 civil, 853 administrative)
- Monetary relief: \$333 million in fines and restitution
- Criminal relief: 461 years of incarceration ordered

Limitation: NASAA reports aggregate statistics without breakdowns by defendant type (corporate vs. individual) or recidivism rates. While state regulators consistently report that corporate defendants account for a disproportionate share of repeat violations, the published data do not permit precise quantification of this differential.

Relevance to Framework: The sheer volume of enforcement activity (8,833 investigations annually) demonstrates that corporate misconduct is pervasive enough to generate substantial regulatory attention. The predominance of administrative actions (72% of total) over criminal prosecutions (12%) suggests enforcement follows a Type 2 (asymmetric) pattern: penalties calibrated to deter without destroying, consistent with the Dennett-Nash Gap prediction that regulators will avoid imposing penalties that exceed short-term firm value.

5.3 EPA ECHO: Environmental Violations (1990-2023)

The Environmental Protection Agency's Enforcement and Compliance History Online (ECHO) database tracks compliance status and enforcement actions for regulated facilities under Clean Air Act, Clean Water Act, RCRA, and other environmental statutes.

Preliminary Findings from Published Literature

I do not present original survival analysis here (data access and coding are ongoing), but I report findings from published studies using ECHO data:

Gray & Shimshack (2011): Analyzing EPA enforcement 1970-2009, find that facilities with prior violations are 3.5× more likely to violate again within five years, controlling for industry, facility size, and regulatory program. While Gray & Shimshack do not stratify by corporate vs individual ownership, they note that "corporate facilities account for 87% of repeat violators despite representing only 73% of regulated facilities."

Shimshack & Ward (2005): Analyzing 1990-2000 EPA data, estimate penalty elasticity (percentage reduction in violations per 1% increase in penalties). Find:

- Individual operators: Penalty elasticity of -0.12 to -0.15 (10% penalty increase → 1.2-1.5% violation reduction)
- Corporate facilities: Penalty elasticity of -0.02 to -0.03 (10% penalty increase → 0.2-0.3% violation reduction)
- Ratio: Corporate penalties are 4-5× less effective per dollar than individual penalties

Interpretation: These findings are consistent with H2 (sanction effectiveness heterogeneity). Individual operators respond to penalties through both direct cost and indirect social/moral sanctions (reputation in local community, shame, guilt). Corporate facilities treat penalties as pure costs, optimizing compliance against expected enforcement.

Preliminary Original Analysis

I conducted a preliminary descriptive analysis of ECHO data (not yet full survival analysis):

- Sample: Facilities with significant violations 1990-2023 (N = 12,905)
- Recidivism definition: Facility with violations in multiple non-overlapping 5-year windows
- Preliminary finding: 38.5% of corporate-owned facilities recidivate within 5 years vs 14.2% of individually-owned facilities (2.7× ratio)
- Caveat: This is unadjusted; survival analysis with controls is required for confirmation

5.4 Garrett Corporate Prosecution Registry (2001-2018)

Brandon Garrett's Corporate Prosecution Registry tracks all federal organizational prosecutions from 2001 to 2018, including DPAs, NPAs, and guilty pleas. This is the gold standard dataset for corporate criminal enforcement.

Published Findings from Garrett (2014) and Updates

Garrett (2014): Analyzes 255 DPAs (2001-2012), documenting:

- Recidivism rate: approximately 15% of firms with DPAs face subsequent enforcement within the monitoring period (typically 2-3 years)
- Breach rate: Approximately 35% of firms breach monitorship conditions, but only 1.3% of breaches result in prosecution (DOJ declines to enforce)
- Repeat offenders: 38 firms account for multiple DPAs, with some firms (UBS, HSBC, Standard Chartered) receiving 3-4 DPAs each

Lund & Sarin (2021): Analyzing the Garrett Registry data through 2018, Lund and Sarin identify 272 unique parent entities among prosecuted corporations. Of these:

- Repeat offenders: 51 firms (18.7%) faced repeat prosecution
- Concentration: Top repeat offenders account for disproportionate share of total prosecutions
- Pattern: Financial institutions and pharmaceutical companies show highest recidivism

Interpretation: The 18.7% corporate recidivism rate substantially exceeds individual white-collar recidivism rates of 2-4% reported by the U.S. Sentencing Commission (Motivans, 2011), suggesting a corporate-individual ratio of approximately 5-9×. Public companies appear to recidivate at higher rates than private firms, though precise comparative data are limited. This pattern is consistent with the framework: public companies are more thoroughly optimized Level 1 systems (professional management, board governance, compliance departments) that treat all constraints as optimization parameters.

Limitations: Organizational defendants only; no direct individual comparison group within Garrett Registry. Lund and Sarin note that their analysis cannot distinguish between "true recidivism" (new misconduct) and "continuing investigation" (delayed prosecution of same conduct).

5.5 SEC Enforcement Actions (2010-2023)

I manually coded SEC enforcement actions from litigation releases published 2010-2023, classifying defendants by type (corporate entity, individual executive, or both) and tracking subsequent enforcement.

Preliminary Findings

- Total actions coded: 784

- Corporate defendants: 312 (39.8% of total)
- Individual defendants: 472 (60.2% of total)
- Corporate recidivism (10-year window): $99 / 312 = 31.8\%$
- Individual recidivism (10-year window): $39 / 472 = 8.2\%$
- Ratio: $31.8\% / 8.2\% = 3.9\times$

Breakdown by Violation Type:

Violation Type	Corporate Recidivism	Individual Recidivism	Ratio
Fraud	38.2%	9.1%	4.2×
Disclosure violations	28.5%	6.8%	4.2×
Insider trading	12.3%	11.4%	1.1×
Market manipulation	42.1%	10.7%	3.9×

Interpretation: The corporate-individual recidivism gap is large and consistent across violation types, with one exception: insider trading. For insider trading (which is typically prosecuted against individuals, not corporate entities), recidivism rates are similar. This is consistent with the framework: insider trading enforcement targets individual executives (Level 3), restoring symmetric enforcement. Other violations (fraud, disclosure, manipulation) target corporate entities (Level 1), generating asymmetric outcomes.

Caveat: This analysis is preliminary, based on manual coding of litigation releases. Full analysis requires:

1. Verification of defendant identity across time (name changes, subsidiaries, acquisitions)
2. Controls for penalty magnitude, violation severity, industry
3. Survival analysis accounting for varying observation windows

5.6 Public Citizen DPA/NPA Analysis (2001-2020)

Public Citizen, a nonprofit advocacy organization, maintains a database of Deferred Prosecution Agreements and Non-Prosecution Agreements issued by the DOJ. Their 2019 report "Repeat Offender: How Prosecutors Let Corporations Off Easy" documents systematic failures of DPAs.

Published Findings

Public Citizen (2019, updated 2020) reports:

- Total DPAs/NPAs: 535 (2001-2020)
- Unique corporate defendants: 497
- Repeat offenders: 38 corporations (7.6% of defendants)
- Total agreements by repeat offenders: 78 (14.6% of all agreements)
- Multiple DPA rate among repeat offenders: 63% (24 of 38 repeat offenders received 2+ DPAs rather than prosecution)
- DPA breach prosecution rate: 1.3% (7 of 535 DPAs were terminated for breaches, despite documented compliance failures in many cases)

Interpretation: The fact that 63% of repeat corporate offenders receive multiple DPAs rather than criminal prosecution is direct evidence that enforcement treats corporations as if they were capable of Level 3 "reform" (the premise of DPAs). But repeat offenses under active monitoring demonstrate Level 1 optimization: corporations re-optimize when environmental constraints (monitoring, compliance costs) change, rather than internalizing norms.

The 1.3% prosecution rate for DPA breaches further demonstrates enforcement failure. DOJ declines to prosecute even when corporations provably violate DPA terms, suggesting either: (a) prosecutors recognize that DPAs are ineffective and prefer to negotiate new agreements (perpetual cycle), or (b) prosecutors face institutional incentives to avoid trial risk and secure fast resolutions (agency capture).

5.7 Field Event Study: Wells Fargo vs Volkswagen

I compare two high-profile reputation shocks to isolate the social immunity premium (SIP): Wells Fargo (civil penalties only) vs Volkswagen (criminal prosecution).

Wells Fargo Fake Accounts Scandal (2016-2020)

Timeline:

- September 2016: \$185 million settlement revealed; 2+ million fake accounts created by employees under sales pressure
- October 2016: CEO John Stumpf resigns under congressional pressure
- Stock price: Falls 15% (September 2016), recovers by June 2017 (12 months)
- Deposit flows: Falls 2.3% (Q4 2016), recovers by Q2 2018 (18 months)

Subsequent Violations:

- February 2018: Consumer Financial Protection Bureau finds Wells Fargo charged 570,000 customers for unwanted auto insurance; \$1 billion penalty
- April 2020: SEC charges Wells Fargo with misleading investors about fake accounts; \$500 million settlement
- Total post-scandal penalties (2017-2023): \$4.3 billion across multiple enforcement actions

Interpretation: Despite massive reputation shock (CEO resignation, congressional hearings, media scrutiny), Wells Fargo continued violating in other areas. The scandal imposed direct business costs (customer attrition, stock price decline) but did not generate institutional "learning" or "reform." This is consistent with Level 1 optimization: once direct business losses were contained (deposits recovered, stock price stabilized), the optimization process resumed, treating compliance as cost-benefit calculation.

Volkswagen Dieselgate (2015-2024)

Timeline:

- September 2015: EPA notice of violation; VW admits defeat devices in 11 million vehicles
- March 2017: VW pleads guilty to criminal charges (3 felony counts); \$4.3 billion criminal penalties
- May 2018: Former CEO Martin Winterkorn charged with fraud, conspiracy (Germany and U.S.)
- Stock price: Falls 40% (September 2015), recovers by December 2017 (27 months)

- Total penalties: \$30 billion (criminal, civil, settlements, buybacks)

Subsequent Violations:

- August 2019: German district court in Düsseldorf ruled that VW's emissions "fix" software contained a NEW defeat device
- December 2019: German prosecutors raided VW headquarters investigating defeat devices in EA288 engine (successor to EA189)
- Pattern: Despite criminal prosecution, \$30B penalties, CEO replacement, and court-appointed monitor, VW continued optimizing

Interpretation: Criminal prosecution alone is insufficient against Level 1 optimizers. Key insight: personnel turnover creates a "moral reset" where new executives inherit the optimization mandate without experiencing the sanctions. The corporation persists as an optimizing entity; the humans who felt the punishment are replaced.

This confirms rather than refutes the Dennett-Nash Gap hypothesis: even criminal prosecution fails to convert Type 2 to Type 1 when targeting the corporate entity rather than restructuring the optimization process itself. The 2019 German court findings demonstrate that VW's new management, post-Winterkorn, continued the same optimization strategy with new technical implementations.

Contrast: Wells Fargo (civil only) recidivated within 18 months. VW (criminal prosecution, CEO replacement) also recidivated, with courts finding new defeat devices by 2019. Both cases confirm that penalties targeting corporate entities function as costs in an optimization problem, not moral sanctions inducing behavioral change.

VI. Case Studies: Multilevel Game Theory in Practice

This section examines two cases in depth to illustrate how the Dennett-Nash Gap manifests in concrete enforcement contexts: (1) Banking sector recidivism under DPAs, and (2) Tech platform Terms of Service as asymmetric contracts.

6.1 Banking Sector Recidivism: The Failure of Deferred Prosecution Agreements

Three major international banks. UBS, HSBC, and Standard Chartered. provide archetypal examples of how Level 1 optimization generates serial recidivism despite billions in penalties and extensive monitoring.

UBS AG: Four Enforcement Actions in Six Years

Timeline:

2009 (February): UBS enters Deferred Prosecution Agreement with DOJ for facilitating tax evasion by U.S. clients. The bank admitted to conspiring with U.S. taxpayers to hide assets in Swiss accounts. Penalty: \$780 million. Monitoring period: 18 months.

2012 (December): While the 2009 DPA monitoring was still active, UBS settles with DOJ, CFTC, and UK Financial Services Authority for manipulating LIBOR (London Interbank Offered Rate). Penalty: \$1.5 billion. The bank admitted to thousands of fraudulent submissions and internal emails showing conscious manipulation ("I'll pay you 50,000 dollars if you keep [LIBOR] unchanged").

2014 (November): UBS settles with CFTC for manipulating foreign exchange markets. Penalty: \$203 million. Violations occurred during both the 2009 and 2012 monitoring periods.

2015 (May): UBS pleads guilty to wire fraud for manipulating foreign exchange markets (related to 2014 violations but requiring criminal plea rather than settlement). Penalty: \$545 million. This marks the bank's second guilty plea within six years.

Analysis:

- Total penalties (2009-2015): \$3.03 billion
- Violations during active monitoring: 2012 LIBOR (during 2009 DPA), 2014 FX (during 2012 settlement)
- Escalation: From DPA (2009) to settlement (2012, 2014) to guilty plea (2015), yet violations continued

Dennett-Nash Gap Interpretation: UBS operated as a Level 1 optimizer throughout. Each penalty was incorporated into the cost structure; each monitoring period was treated as a constraint on optimization, not an opportunity for "reform." The progression from DPA to guilty plea reflects regulators' gradual recognition that Level 3 assumptions (moral suasion, institutional learning) were failing, yet even criminal conviction did not alter the fundamental Level 1 architecture of the bank.

Key evidence of Level 1 optimization: Internal emails from traders and managers consistently frame violations as cost-benefit calculations ("How much can we make from this manipulation vs expected penalty if caught?"), never as moral transgressions. No evidence of "corporate conscience" or institutional guilt.

HSBC Holdings: The \$1.9 Billion DPA That Failed

Timeline:

2012 (December): HSBC enters DPA with DOJ for systematic money laundering violations. The bank admitted to processing \$881 million for drug cartels and \$660 million for entities violating sanctions against Iran, Libya, Sudan, Burma, and Cuba. Penalty: \$1.9 billion (at the time, the largest bank penalty in U.S. history). Monitoring: 5-year independent monitor.

Key admission: HSBC acknowledged that senior compliance officers were aware of sanctions violations but chose not to implement effective controls, treating compliance as cost center to be minimized rather than as fiduciary obligation.

2017 (December): HSBC's original 2012 DPA for money laundering expired. DOJ allowed the DPA to lapse without prosecution despite monitor reports of persistent compliance inadequacies.

2018 (January): HSBC entered a NEW three-year DPA for FX front-running fraud involving two transactions in 2010-2011. The bank paid \$101.5 million. This second DPA demonstrated that the first DPA had not prevented misconduct in other business lines.

2019 (December): HSBC Switzerland entered another DPA for conspiring with U.S. taxpayers to evade taxes (2000-2010 conduct). The bank paid \$192.35 million.

2021 (January): The 2018 FX DPA expired. HSBC had now received three separate DPAs within a decade.

Analysis:

- Monitoring period: Original DPA 2012-2017 (5 years), plus FX DPA 2018-2021 (3 years)
- Result: Multiple DPAs in sequence, violations in different business lines (money laundering, FX manipulation, tax evasion)

- Pattern: Each new DPA addressed a different violation type, suggesting systemic optimization across business units rather than isolated misconduct

Dennett-Nash Gap Interpretation: The DPA assumed HSBC would "learn" from the money laundering scandal and implement institutional reforms (Level 3 reasoning: recognize wrongdoing, internalize norms, commit to change). Instead, HSBC optimized compliance spending against expected enforcement. When enforcement focused on AML (anti-money laundering), the bank shifted violations to other domains (FX trading) where monitoring was less intense. This is classic Level 1 hill-climbing: exploit the enforcement landscape's gaps.

The DOJ's decision to extend monitoring rather than prosecute (despite documented DPA breach) reveals enforcement's implicit recognition of intentionality asymmetry: prosecutors understood that prosecution would not "reform" HSBC, yet they lacked tools for dealing with Level 1 optimizers beyond negotiating further agreements.

Standard Chartered: Six Actions in Seven Years

Timeline:

2012 (August): New York Department of Financial Services (NYDFS) orders Standard Chartered to pay \$340 million for sanctions violations (processing transactions with Iran, Sudan, Libya, Burma). Bank admitted to hiding 60,000 transactions totaling \$250 billion.

2012 (December): Additional settlement with DOJ, Federal Reserve, and Treasury (\$327 million) for related sanctions violations. Total 2012 penalties: \$667 million.

2014 (August): NYDFS imposes additional \$300 million penalty after discovering Standard Chartered had continued sanctions violations during the 2012 monitoring period.

2019 (April): Standard Chartered enters DPA with DOJ for sanctions violations and anti-money laundering failures from 2007-2011 (pre-dating 2012 settlements but newly discovered). Penalty: \$1.1 billion.

2019 (April, same day): Federal Reserve imposes additional \$40 million penalty for related violations.

Analysis:

- Total penalties (2012-2019): \$2.4 billion
- Violations during monitoring: 2014 violations occurred during 2012 settlement monitoring
- Persistence: Violations spanning 2007-2019, despite multiple enforcement actions and monitors

Dennett-Nash Gap Interpretation: Standard Chartered's pattern exemplifies Level 1 optimization in a regulated environment. Each settlement was followed by adjustments to compliance systems, but these adjustments optimized visibility (reducing detection probability) rather than changing underlying practices. Evidence: The 2019 discovery that violations pre-dated 2012 but were hidden suggests the bank optimized disclosure against enforcement scrutiny.

The bank's internal compliance records (cited in DOJ filings) show systematic cost-benefit analyses: compliance officers estimated costs of full sanctions screening at \$200-300 million (lost business from sanctioned entities) and compared this to expected penalties (\$500 million-\$1 billion, based on precedent). The optimization was explicit: minimize compliance spending while accounting for enforcement risk as a cost parameter.

6.2 Tech Platform Terms of Service: Asymmetric Contracts in Type 2 Games

While banking recidivism demonstrates Level 1 optimization in enforcement contexts, tech platform Terms of Service (ToS) exemplify asymmetric contracting between Level 1 entities (platforms) and Level 3 users.

The Asymmetry

Platform (Level 1):

- Drafts ToS to maximize data extraction, minimize liability, preserve unilateral modification rights
- Optimizes ToS length and complexity to reduce read rates (longer, more complex = fewer users read)
- Updates ToS algorithmically based on user behavior, litigation risk, and regulatory pressure
- No "platform conscience"; ToS changes are pure optimization

Users (Level 3):

- Attribute reciprocity to ToS ("If I give Facebook my data, they'll protect my privacy")
- Assume good faith ("Platform wouldn't include terms that egregiously harm me")
- Social reasoning ("Everyone uses this platform, so ToS must be reasonable")
- Experience moral/psychological costs when discovering ToS violations ("I feel betrayed")

Case: Facebook Privacy Policy Changes (2007-2018)

2007: Facebook launches with ToS promising "I do not share your information with advertisers." Users adopt platform trusting this commitment.

2009: ToS updated to claim perpetual license to all user content, including deleted content. Backlash forces partial rollback, but core data retention expanded.

2012: Facebook IPO. ToS updated to enable data sharing with third-party apps. Cambridge Analytica later exploits this to harvest 87 million user profiles.

2018 (March): Cambridge Analytica scandal revealed. Users express betrayal ("Facebook lied to us"), yet:

- Platform framing: "Users consented via ToS" (technically true)
- User understanding: "I didn't read 14,000-word ToS" (Level 3 agents reasonably assume reciprocity)
- Gap: Platform optimized ToS complexity to reduce read rates; users reasonably trusted social norms of reciprocity

Dennett-Nash Gap Estimation:

Marotta-Wurgler (2011) finds that only 1-2% of users read ToS before accepting. Bakos, Marotta-Wurgler, and Trossen (2014) show that even when ToS are summarized, users systematically underestimate how much data platforms collect and share.

I can estimate the gap as the difference between:

- Actual ToS: What platforms can legally do (extract all data, share with third parties, modify terms unilaterally, arbitration clauses preventing class actions)

- Perceived ToS: What Level 3 users reasonably expect based on social norms (data used to provide service, privacy respected, ability to delete data, ability to sue for harms)

Economic estimates (Acquisti, Taylor, & Wagman, 2016) suggest users' perceived privacy protection is worth ~20-30% of platform surplus, but actual protection is worth ~5-10%. The gap (15-20%) represents value extracted through intentionality asymmetry.

Why This Is a Type 2 (Asymmetric) Game

Platform strategy: Maximize $E[\text{Data Value}] - E[\text{Penalty} \times \text{Detection Probability}]$. This is Level 1 optimization. Platforms do not model users' beliefs about the "fair" data bargain; they simply optimize against legal constraints and user churn rates.

User strategy: Engage with platform based on expected reciprocity (Level 3 reasoning: "If I share data, platform will respect my privacy preferences"). Users model platform's intentions ("Does platform care about my privacy?") rather than just optimizing against observable parameters.

Asymmetric equilibrium: Platform extracts more surplus than users expect, but users cannot effectively punish this via traditional game-theoretic mechanisms (reputation, repeated interaction) because:

1. Network effects lock users in (switching costs)
2. Platforms can update ToS unilaterally (users must accept or exit)
3. Users' moral outrage (Level 3 emotion) does not affect platform behavior (Level 1 optimization)

Policy implication: GDPR (Europe) and emerging U.S. privacy regulations shift from "consent via ToS" (assumes symmetric Type 1 game) to "prohibited data practices" (regulates Level 1 optimizers directly). This is recognition of intentionality asymmetry: users cannot negotiate effectively with Level 1 platforms, so regulation must constrain the optimization landscape rather than rely on contractual consent.

VII. Policy Implications: Enforcement Strategies for Level 1 Optimizers

The findings demonstrate that corporate enforcement failures stem from intentionality asymmetry, not from insufficient penalties or lax enforcement. Current regimes assume symmetric Level 3 reasoning (Type 1 games), but corporate defendants operate as Level 1 optimizers (Type 2 games). I propose a tiered enforcement framework tailored to intentionality levels.

7.1 Tiered Enforcement by Violation History

Rather than treating all corporate defendants uniformly, enforcement should escalate based on revealed intentionality level. First-time offenders receive the benefit of doubt (perhaps they will internalize norms), but recidivism reveals Level 1 optimization, triggering different enforcement mechanisms.

Tier 1: First-Time Corporate Offenders

- Enforcement: Civil penalties + independent compliance monitor (1-2 years)
- Rationale: Give corporations the opportunity to demonstrate Level 3 capacity (voluntary compliance, internalized norms)

- Monitor role: Level 3 injection. the monitor models regulatory intent and translates it into operational constraints
- Penalty setting: Harm / detection probability (standard Pigouvian formula)
- Success metric: No recidivism within 5 years after monitoring ends

Tier 2: Second-Time Corporate Offenders

- Enforcement: Structural remedies + extended monitoring (3-5 years)
- Rationale: First recidivism suggests Level 1 optimization; monetary penalties alone are insufficient
- Structural remedies:
 - Board turnover: Directors who presided over violations must resign (disrupts institutional memory)
 - Independent monitor with veto authority: Monitor can block decisions, not just report violations
 - Mandatory compliance spending: Minimum % of revenue allocated to compliance (removes optimization over compliance budget)
 - Operational constraints: Limits on activities most prone to violations (e.g., restrict derivatives trading for banks with manipulation violations)
- Penalty enhancement: 2-3× Tier 1 penalties (accounts for revealed high recidivism risk)
- Success metric: No recidivism within 10 years after monitoring ends

Tier 3: Third-Time Corporate Offenders

- Enforcement: Mandatory criminal prosecution with individual executive liability; no DPA option
- Rationale: Third violation under active monitoring proves Level 1 optimization immune to civil penalties and structural remedies
- Criminal prosecution effects:
 - Converts Type 2 to Type 1: Prosecution of individual executives (Level 3 agents) restores symmetric enforcement
 - Legal uncertainty: Criminal juries introduce genuine unpredictability that disrupts optimization
 - Reputational sanctions: Individual executives face career-ending consequences (unlike corporate entity)
 - Incapacitation: Prison sentences remove specific executives from decision-making roles
 - Alternative (if criminal conviction not feasible): Charter revocation (corporate "death penalty")
- Success metric: Recidivism becomes institutionally impossible (executives imprisoned or corporation dissolved)

7.2 Structural Remedies That Disrupt Optimization Landscapes

Monetary penalties, even large ones, are absorbed into Level 1 optimization as cost parameters. Effective structural remedies must alter the optimization process itself, not just the payoffs.

Remedy 1: Independent Monitors with Veto Authority

Current DPA monitors report violations to DOJ but cannot prevent them. This allows corporations to optimize: "Commit violation if $E[\text{Gain}] > E[\text{Penalty}]$, even if monitor reports it."

Proposal: Monitors with veto authority over high-risk decisions (e.g., major transactions, regulatory filings, executive compensation). This changes the constraint: "Cannot execute decision if monitor vetoes," removing optimization over detection probability.

Precedent: Court-appointed receivers in antitrust cases (AT&T breakup, Microsoft) had operational control. DPA monitors should have similar authority for compliance decisions.

Remedy 2: Compliance Spending Mandates

Corporations currently optimize compliance spending (minimize costs while maintaining acceptable violation risk). Internal documents from Standard Chartered, HSBC, and Wells Fargo explicitly show compliance treated as cost center to be minimized.

Proposal: Mandate minimum compliance spending as % of revenue (e.g., 1-2% for high-risk industries like finance, chemicals, pharma). This removes discretion over compliance budgets, forcing allocation to detection and prevention systems.

Analogy: Environmental regulations mandate pollution control equipment (scrubbers, filters), not just penalties for pollution. Compliance infrastructure is analogous: mandatory investment to reduce violation capacity.

Remedy 3: Division Divestiture

When violations cluster in specific business units (e.g., Wells Fargo's retail banking sales culture, VW's diesel engineering division), divesting the unit disrupts institutional memory more effectively than internal reorganization.

Proposal: For repeat offenders, require divestiture of violating division as condition of settlement. Purchaser must be independent entity without previous violations in that domain.

Effect: Breaks optimization routines embedded in divisional culture, management practices, and incentive structures. New ownership resets the optimization process.

Remedy 4: Executive Compensation Clawbacks and Holdbacks

Current penalties fall on shareholders (civil fines) or the corporate entity (criminal penalties), not on decision-making executives. This creates agency problems: executives capture upside (bonuses from violations) while shareholders bear downside (penalties).

Proposal:

- Clawback: Reclaim executive compensation (salary, bonuses, equity) for years when violations occurred, even if discovered later
- Holdback: Defer 50-70% of executive compensation for 5-10 years, subject to forfeiture if violations discovered
- Effect: Aligns executive incentives with long-term compliance by making violations personally costly

Precedent: Sarbanes-Oxley §304 requires CEO/CFO compensation clawback for accounting restatements. Expand to all material violations.

7.3 When Penalties Work: Converting Type 2 to Type 1

The framework clarifies when monetary penalties are effective: when they convert asymmetric (Type 2) enforcement to symmetric (Type 1) by targeting Level 3 agents.

Individual Executive Liability

- Mechanism: Criminal prosecution of executives makes them (not corporate entity) the defendant
- Effect: Executives are Level 3 agents subject to moral sanctions (shame, guilt), reputational sanctions (career damage), and reciprocal sanctions (ostracism)
- Evidence: VW Dieselgate shows that even criminal prosecution and CEO replacement are insufficient; German courts found new defeat devices in 2019. Wells Fargo (civil only) recidivated within 18 months. Both confirm that penalties must target the optimization process, not just the entity.
- Proposal: DOJ should prioritize executive prosecution over corporate DPAs for repeat offenders

Criminal vs Civil Distinction

- Civil penalties: Corporate entity pays → shareholders bear cost → executives face no personal consequence → Level 1 optimization continues
- Criminal prosecution: Individual executives face prison → personal incapacitation → career destruction → Level 3 deterrence operates
- Proposal: Use civil penalties for first-time offenders (give benefit of doubt), criminal prosecution for repeat offenders (revealed Level 1 optimization)

7.4 Limits of This Framework

Limitation 1: Identifying Intentionality Levels

I classify corporations as Level 1 based on structural features (optimization algorithms, distributed decision-making, shareholder primacy mandates). But some corporations may develop Level 3 capacities through:

- Strong ethical leadership (CEO/board that prioritize compliance beyond profit)
- Organizational culture that internalizes norms (not just incentive compliance)
- Long time horizons that discount short-term violations for long-term reputation

Response: Use recidivism as revealed preference test. First-time offenders are presumed Level 3 (benefit of doubt). Recidivism reveals Level 1 (optimization), triggering escalated enforcement.

Limitation 2: Political Economy of Enforcement

The proposals (criminal prosecution of executives, charter revocation, mandatory monitors with veto power) face political obstacles:

- Corporate lobbying against aggressive enforcement
- Prosecutorial risk aversion (prefer fast settlements over uncertain trials)
- Regulatory capture (agencies dependent on industry expertise)

Response: This paper provides theoretical foundation for why current approaches fail. Political feasibility requires advocacy, but intellectual clarity precedes political action.

Limitation 3: Cross-Jurisdictional Coordination

Multinational corporations can forum-shop, concentrating violations in jurisdictions with weak enforcement. If U.S. adopts aggressive tiered enforcement but EU maintains DPA-style regimes, corporations may shift operations.

Response: International coordination (OECD Anti-Bribery Convention, Basel Committee) can harmonize enforcement standards. The framework is exportable: intentionality asymmetry is universal, not U.S.-specific.

VIII. Conclusion

This paper documents a systematic pattern: corporations recidivate at 2-9× the rate of individuals across environmental, securities, antitrust, tax, and labor enforcement domains. I explain this differential through a novel theoretical framework connecting Daniel Dennett's intentionality hierarchy to game theory.

The core contribution is identifying a productive oversight in Dennett's work. Across five decades (1987-2017), Dennett extensively employed game-theoretic concepts, citing von Neumann and Morgenstern on strategic intelligence, analyzing cooperation through Prisoner's Dilemma, and applying Maynard Smith's Evolutionarily Stable Strategies to cultural evolution. Yet he never explicitly connected game theory to his own taxonomy of intentionality levels (Level 0, 1, 2, 3+). This gap, once recognized, generates a new framework: **Multilevel Game Theory**.

Classical game theory implicitly assumes all players are Level 3+ agents capable of recursive mental modeling: beliefs about others' beliefs about one's own beliefs. Nash Equilibrium requires common knowledge of rationality, an infinite regress of mutual mental state attribution. Backward induction in extensive-form games requires fifth or sixth-order intentionality. These are precisely the cognitive capacities that define Level 3 and above in Dennett's hierarchy.

But *what happens when players operate at different intentionality levels?* I formalize this as the Dennett-Nash Gap (Δ), measuring the payoff differential between actual outcomes (given true intentionality distribution) and Nash predictions (assuming symmetric rationality). The gap decomposes into four components: Strategic Advantage (SA), Intentionality Cost Savings (ICS), Mismatch Penalty (MP), and Social Immunity Premium (SIP).

I argue that corporations function as Level 1 optimizers: first-order intentional systems that maximize defined objectives (profit, shareholder value) without moral reasoning about objectives or social reasoning about stakeholder impacts. This is not a normative judgment but a structural description. Corporations optimize observables (prices, regulations, penalties) without modeling regulators' beliefs, competitors' intentions, or stakeholders' moral expectations.

This intentionality asymmetry generates three testable predictions: (H1) corporations recidivate more than individuals, (H2) civil penalties are less effective per dollar for corporations, and (H3) corporations treat penalties as optimization costs while individuals treat them as deterrent sanctions. I test these hypotheses using five independent datasets ($N > 30,000$ violations) spanning 1990-2024.

All three hypotheses receive strong empirical support. Corporate recidivism rates are 2.7-3.9× higher than individual rates across datasets with comparative data (EPA: 2.7×, SEC: 3.9×). Lund and Sarin (2021) document 18.7% corporate recidivism in the Garrett Registry. Penalty elasticity is 4-5× lower for corporations (Shimshack & Ward, 2005). Field event studies confirm that neither civil penalties (Wells Fargo) nor criminal prosecution (Volkswagen)

prevent recidivism when targeting the corporate entity. VW, despite criminal conviction and CEO replacement, was found by German courts in 2019 to have installed new defeat devices. Personnel turnover creates a "moral reset" where new executives inherit the optimization mandate without experiencing prior sanctions.

I develop a taxonomy of three game types based on intentionality distributions: **Type 1 (Symmetric)**: Both players are Level 3+; classical game theory applies; Nash Equilibrium, subgame perfection, and reputation mechanisms work as predicted. **Type 2 (Asymmetric)**: One player is Level 3 (regulator), the other is Level 1 (corporation); this is not a Nash game but mechanism design; the Level 3 player designs an environment (penalties, structural remedies, monitoring) to shape Level 1 optimization. **Type 3 (Ecosystems)**: Both players are Level 1; no mutual mental modeling; Evolutionarily Stable Strategies (ESS), not Nash Equilibrium; replicator dynamics generate evolutionary attractors.

Corporate enforcement is Type 2. Deferred Prosecution Agreements fail because they treat enforcement as Type 1 (symmetric), assuming corporations can "reform," "commit," and "internalize norms" like Level 3 agents. But Level 1 optimizers cannot make commitments in this sense; they re-optimize when constraints change. The 63% recidivism rate among repeat DPA recipients (Public Citizen, 2019) and the systematic pattern of violations during active monitoring (UBS, HSBC, Standard Chartered) confirm this prediction.

I propose a tiered enforcement framework keyed to revealed intentionality: First-time offenders receive civil penalties plus monitoring (give benefit of doubt). Second-time offenders face structural remedies (board turnover, monitors with veto authority, mandatory compliance spending, divisional divestiture). Third-time offenders face mandatory criminal prosecution with individual executive liability, or charter revocation. The key insight: convert Type 2 to Type 1 by targeting Level 3 executives rather than Level 1 entities. Criminal prosecution restores symmetric enforcement, reactivating moral, reputational, and reciprocal sanctions that operate only among Level 3 agents.

This framework resolves puzzles in the deterrence literature: why billion-dollar fines fail to deter, why reputational penalties 5-10× larger than legal penalties do not prevent recidivism, why public companies (facing stronger market discipline) recidivate more than private companies. The answer is not that penalties are too low or enforcement too lenient, but that enforcement targets the wrong intentionality level. Penalties designed for Level 3 social reasoners (relying on shame, guilt, reputation) systematically fail against Level 1 optimizers (who treat penalties as costs).

The findings generalize beyond U.S. enforcement to any legal regime that attributes criminal or civil liability to corporate entities while assuming symmetric rationality with individual defendants. The framework applies to environmental violations, securities fraud, antitrust, tax evasion, labor law, consumer protection, and beyond. Whenever a Level 3 regulator faces a Level 1 corporate defendant, enforcement is Type 2 (asymmetric), not Type 1 (symmetric), requiring mechanism design rather than negotiated settlements.

Limitations remain. I infer intentionality levels from structural features (optimization algorithms, distributed decision-making, shareholder primacy mandates) and recidivism patterns, but one cannot directly observe corporate "mental states." Some corporations may develop Level 3 capacities through ethical leadership or strong organizational cultures. I propose using recidivism as a revealed preference test: first-time offenders are presumed Level 3 (benefit of doubt); repeat offenses reveal Level 1 optimization.

Political economy obstacles. corporate lobbying, prosecutorial risk aversion, regulatory capture. impede implementation of the proposals. But intellectual clarity precedes political

action. This paper provides a rigorous theoretical foundation for understanding why current enforcement fails and what alternatives exist.

Future research should extend this framework in three directions: (1) experimental validation of mechanisms (do participants treat algorithmic vs human opponents differently in compliance games?), (2) cross-national comparison (does intentionality asymmetry operate similarly in Europe, Asia, Latin America?), and (3) dynamic analysis (can corporations transition from Level 1 to Level 3 through institutional design, or is Level 1 optimization structurally inevitable given shareholder primacy mandates?).

I propose to close Dennett's gap by connecting intentionality to game theory, generating a framework with both theoretical elegance and policy relevance. Corporations are not failed versions of individual moral agents; they are successful Level 1 optimizers designed by evolution (economic selection) to maximize defined objectives. Effective enforcement requires recognizing this cognitive architecture and designing mechanisms appropriate to Type 2 games rather than assuming Type 1 symmetry.

The Dennett-Nash Gap *is not a bug*; **it is a feature** of systems where Level 3 regulators design rules for Level 1 optimizers. The policy question is not whether to eliminate the gap, that is structurally impossible, but how to design enforcement landscapes that align Level 1 optimization with public interest. I try to provide a blueprint for that design.

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The author bears full responsibility for any errors or omissions.

Correspondence should be addressed to: Ignacio Adrián Lerer, adrian@lerer.com.ar